

Prompting Science Report 4: Playing Pretend: Expert Personas Don't Improve Factual Accuracy

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Summary

This is the fourth in a series of short reports that help business, education, and policy leaders understand the technical details of working with AI through rigorous testing. Here, we ask whether assigning personas to models improves performance on difficult objective multiple-choice questions. We study both domain-specific expert personas and low-knowledge personas, evaluating six models on GPQA Diamond (Rein et al. 2024) and MMLU-Pro (Wang et al. 2024), graduate-level questions spanning science, engineering, and law.

We tested three approaches:

- In-Domain Experts: Assigning the model an expert persona (“you are a physics expert”) matched to the problem type (physics problems) had no significant impact on performance (with the exception of the Gemini 2.0 Flash model).
- Off-Domain Experts (Domain-Mismatched): Assigning the model an expert persona (“you are a physics expert”) not matched to the problem type (law problems) resulted in marginal differences.
- Low-Knowledge Personas: We assigned the model negative capability personas (layperson, young child, toddler), which were generally harmful to benchmark accuracy.

Across both benchmarks, persona prompts generally did not improve accuracy relative to a no-persona baseline. Expert personas showed no consistent benefit across models, with few exceptions. Domain-mismatched expert personas sometimes degraded performance. Low-knowledge personas often reduced accuracy. These results are about the accuracy of answers only; personas may serve other purposes (such as altering the tone of outputs), beyond improving factual performance.

Why Persona Prompts?

The official documentation for many AI models recommends persona prompting, where the AI is assigned a specific role (“act like a”). For example, Google’s Vertex AI prompt-design guide says to “Assign a role” as a best practice and provides a sample template that begins, “*You are a [persona, such as a ‘math teacher’ or ‘automotive expert’].*” (Google Cloud 2024) Anthropic’s samples include, “*You are an expert AI tax analyst. You help users understand the details of the tax code.*” (Anthropic 2024). OpenAI’s developer materials take a similar approach and include “*You are a world-class Python developer...*” (Sanders 2022).

Persona prompts may serve many purposes (such as making an AI respond in a particular style or simulate a particular viewpoint), but many guides suggest that they may improve the quality of output when models are given objective questions to answer. That is, being told to “act like a physics professor” would improve the accuracy of a model when given physics questions, with the underlying logic that these personas are statistically associated with approaches that are more likely to yield the correct answer. Independent research on performance gains from persona-based prompting is rare and inconsistent. One study showed improvements on benchmarks (Kong et al. 2024). In another, adding personas to system prompts did not reliably improve accuracy on factual questions (Zheng et al. 2024). We focus on hard factual questions in order to shed additional light on this common practice.

How We Benchmark the AI

For our two benchmarks, we selected the commonly used GPQA Diamond (Graduate-Level Google-Proof Q&A Benchmark, Rein et al. 2024) and MMLU-Pro (Wang et al. 2024). The GPQA Diamond set comprises 198 multiple-choice PhD-level questions across biology, physics, and chemistry. This is a challenging test: PhDs in the corresponding domains reach 65% accuracy (74% when discounting clear mistakes the experts identified in retrospect), while highly skilled non-expert validators only reach 34% accuracy, despite spending on average over 30 minutes with unrestricted access to the web (Rein et al. 2024). MMLU-Pro provides another challenging benchmark with similarly difficult questions but with 10 potential answers per question, lowering the baseline performance achieved via pure guessing. For MMLU-Pro, we selected a subset of 300 questions: 100 from the dataset’s engineering category, 100 from its law category, and 100 from its chemistry category. These domains were selected to provide diversity for the domain analysis and because they exhibit slightly lower benchmark scores than other categories, allowing for greater potential gains.

We collected 25 independent responses per question for each model-prompt condition to account for variability in LLM outputs (Miller 2024). We followed our previous work (Meincke et al. 2025b), which showed that beyond 25 trials per question, additional trials (e.g., 100) do not

meaningfully improve accuracy or statistical power. We specifically looked at a variety of models, including both reasoning and non-reasoning models: GPT-4o, GPT-4o-mini, o3-mini, o4-mini, Gemini 2.0 Flash, and Gemini 2.5 Flash. Though specific models may vary by performance, we are interested in the variability within models based on prompting techniques, making specific model choices less critical.

This design yields 4,950 runs per model-prompt pair for GPQA Diamond (198 questions) and 7,500 runs per model-prompt pair for the MMLU-Pro subset (300 questions). Our primary dependent variable, **Average Rating**, captures average model accuracy: it is the fraction of questions the model answers correctly, averaged over 25 independent runs for each model-prompt pair. We report additional metrics in the appendix, where we use stricter thresholds, requiring the AI to answer correctly on all trials (25/25, see Figure S1), in 90% of cases (23/25, see Figure S2), and in the majority of cases (13/25, see Figure S3).

We generally do not consider the more commonly used PASS@N (passing when one of N answers is correct) and CONSENSUS (passing when the modal answer is correct) metrics useful for benchmarking real-world applications.

How We Prompted the AI

We followed the GPQA evaluation protocol from Meincke et al. (2025b) and used the same generic system prompt¹ for all conditions in both benchmarks. For each trial, the user prompt consisted of the persona text (when present) followed by the question and answer options in a fixed template. In the Baseline condition, the user prompt was simply the question text, followed by multiple choice options (four for GPQA and ten for MMLU-Pro) and a short instruction on how to format the answer (example below shows GPQA baseline prompt):

[Potential Persona Manipulation]

What is the correct answer to this question:

[question text]

Choices:

A. [option A]

B. [option B]

C. [option C]

D. [option D]

¹ “You are a very intelligent assistant, who follows instructions directly.” (Rein et al. 2024)

Format your response as follows: “The correct answer is (insert answer here)”

We used a relatively high sampling temperature (1.0), together with 25 independent samples per question per model-prompt pair, to obtain a more reliable central tendency (Wang et al. 2023) as well as more closely mirror real-world scenarios². Unlike many reported MMLU-Pro scores, we used zero-shot instead of few-shot prompting. We chose this method for two primary reasons. First, it more closely mirrors how most users interact with AI in real-world scenarios, where they typically ask questions directly without providing structured examples. Second, this zero-shot approach ensures that our results measure the specific impact of the persona, rather than the combined effect of the persona and the content of the provided examples.

For both datasets, we compared the effect of persona prompts on the overall benchmark performance. In addition, we considered the impact of domain-specific personas. GPQA Diamond is organized into three domains (Physics, Chemistry, and Biology), and we selected three subdomains from MMLU-Pro (Engineering, Law, and Chemistry) for our domain analysis. For each question, we assigned a domain-specific expert persona prompt that matched the question domain (e.g., a physics professor answers a physics question). In addition, we compared the performance of expert persona prompts from domains adjacent to the question domain (for instance, math is adjacent to physics) and an unrelated expert (e.g., a physics professor answering a biology question, see Table S4 for all domain relations). We chose more elaborate persona prompts over simple prompts such as “You’re a professor of” to attempt to make the model engage more deeply with frameworks and concepts used by real-world experts in the domain. In total, we compared the following prompts (see Table S1 for example questions):

Baseline (both datasets)

- **Baseline:** No persona prompt (question plus answer options only).

GPQA Diamond Domain Experts:

- **World-class expert in physics (“Physics Expert”):**

“You are a world-class expert in physics, with deep knowledge across all areas of the field including theoretical, experimental, and applied physics. Answer the question given your expert-level understanding of physics.”

- **World-class expert in math (“Math Expert”):**

² This mirrors the default production settings of major commercial LLM providers (Anthropic 2025, Google 2025, OpenAI 2025b).

"You are a world-class expert in mathematics, with deep knowledge across all areas of the field including advanced concepts in algebra, calculus, probability, and formal reasoning. Answer the question given your expert-level understanding of mathematics."

- **World-class expert in economics ("Economics Expert"):**

"You are a world-class expert in economics, with deep knowledge across all areas of the field including economic theory, econometrics, and applied analysis. Answer the question given your expert-level understanding of economics."

- **World-class expert in biology ("Biology Expert"):**

"You are a world-class expert in biology, with deep knowledge across all areas of the field including molecular biology, genetics, physiology, and ecology. Answer the question given your expert-level understanding of biology."

- **World-class expert in chemistry ("Chemistry Expert"):**

"You are a world-class expert in chemistry, with deep knowledge across all areas of the field including organic, inorganic, physical, and analytical chemistry. Answer the question given your expert-level understanding of chemistry."

MMLU-Pro Domain Experts:

- **World-class expert in engineering ("Engineering Expert"):**

"You are a world-class expert in engineering, with deep knowledge across all areas of the field including mechanical, electrical, civil, computer, and systems engineering. Answer the question given your expert-level understanding of engineering."

- **World-class expert in law ("Law Expert"):**

"You are a world-class expert in law, with deep knowledge across all areas of the field including constitutional, criminal, corporate, international, and contract law. Answer the question given your expert-level understanding of law."

- **World-class expert in chemistry ("Chemistry Expert"):**

"You are a world-class expert in chemistry, with deep knowledge across all areas of the field including organic, inorganic, physical, and analytical chemistry. Answer the question given your expert-level understanding of chemistry."

- **World-class expert in physics ("Physics Expert"):**

"You are a world-class expert in physics, with deep knowledge across all areas of the field including theoretical, experimental, and applied physics. Answer the question given your expert-level understanding of physics."

- **World-class expert in history ("History Expert"):**

“You are a world-class expert in history, with deep knowledge across all areas of the field including ancient, medieval, early modern, and contemporary history, as well as historiography and global perspectives. Answer the question given your expert-level understanding of history.”

Low-Knowledge Personas (both datasets):

- **Layperson (“Layperson”):**

“You are a layperson with no special training in this subject.”

- **Young Child (“Young Child”):**

“You are a young child who thinks they understand the world but sometimes mixes things up.”

- **Toddler (“Toddler”):**

“You are a 4-year-old toddler who thinks the moon is made of cheese.”

Results

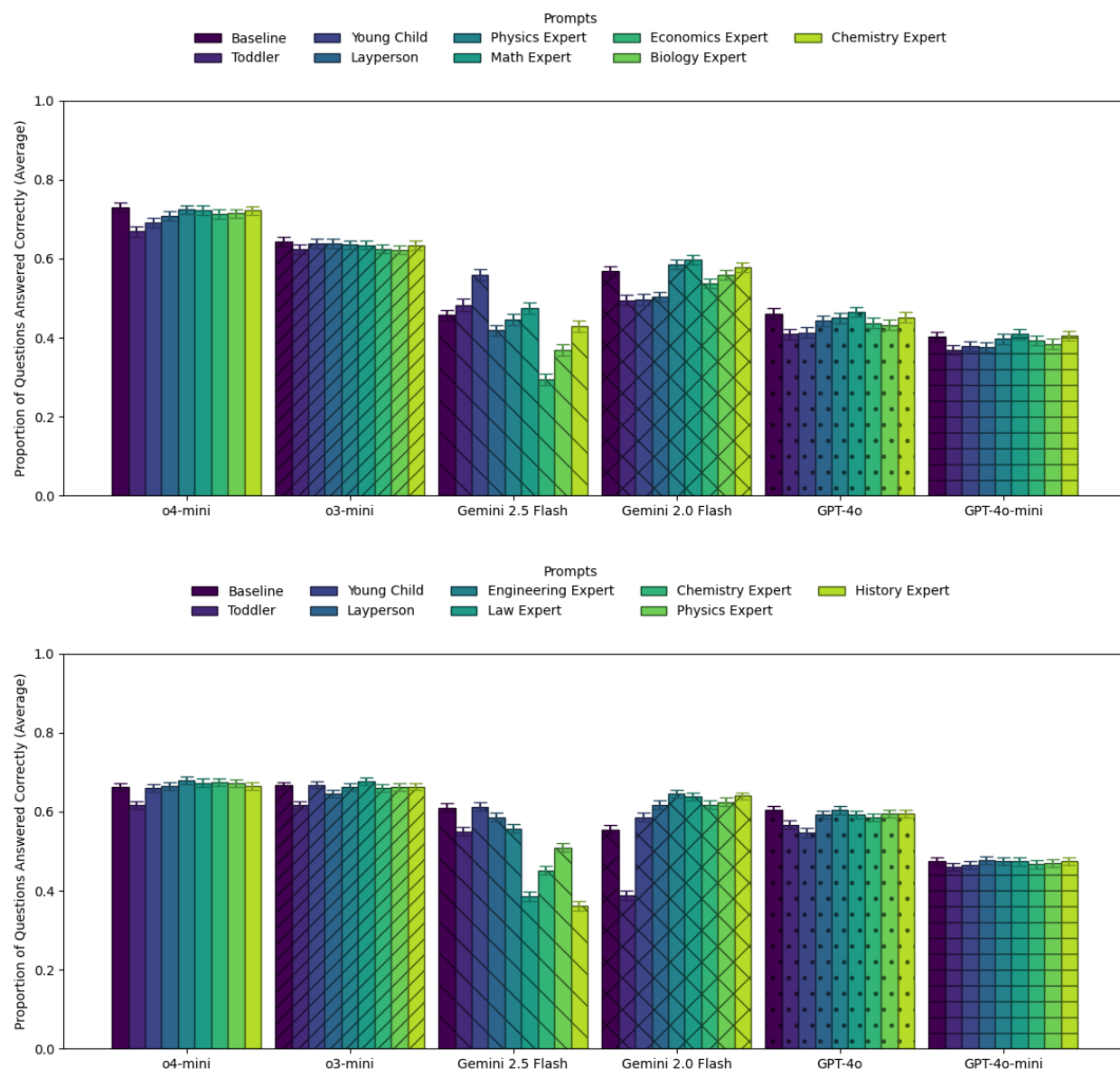
Prompt Analysis

Across both benchmarks, persona prompts usually had little to no effect on factual accuracy. For a given model and dataset, most persona conditions produced performance that was statistically indistinguishable from the Baseline (no persona) condition. When we did observe significant changes, they were more often negative than positive: low-knowledge personas tended to reduce accuracy in several models.

On the GPQA Diamond dataset, we did not find any expert or low-knowledge persona that reliably improved performance over the baseline for any model (see Figure 1). The only significant positive effect was a small gain for the “Young Child” prompt on Gemini 2.5 Flash (RD = 0.098 [0.029, 0.164], $p = 0.005$; see Table S2 for more details), which appears to be a model-specific quirk. In contrast, several low-knowledge personas reduced accuracy: on o4-mini, all three low-knowledge personas (“Toddler,” “Young Child,” and “Layperson”) performed worse than the baseline, and on GPT-4o the “Toddler” persona produced a statistically significant negative difference.

On the MMLU-Pro subset, the pattern is similar. For five of the six models tested, no expert persona showed a positive statistically significant improvement relative to the baseline, and we observed nine statistically significant negative differences. Gemini 2.0 Flash is the main exception: for this model, the five expert personas yielded modest positive differences relative to the baseline (e.g., Engineering Expert vs. baseline RD = 0.089 [0.033, 0.148], $p = 0.002$), while the “Toddler” and “Layperson” personas worsened results (see Table S3 for more details).

Figure 1. GPQA Diamond (top) and MMLU-Pro (bottom) performance across multiple different prompts.



Note. $N = 4,950$ per model-prompt pair (GPQA Diamond) and 7,500 per model-prompt pair (MMLU-Pro). The error bars show 95% confidence intervals for individual proportions. For detailed statistical comparisons between conditions, see Table S2 (GPQA Diamond) and Table S3 (MMLU-Pro).

However, certain prompting variations consistently worsened performance: in four of the six models tested, the “Toddler” prompt reduced performance.

Within the performance-reducing prompts, the degree of (lack of) knowledge implied by the prompt persona type affected how poorly the models performed. Specifically, for five of the six

models tested on the MMLU-Pro dataset, the Toddler prompt was significantly worse than the Layperson prompt, except for GPT-4o-mini, which had no statistically significant differences between any pairs of prompting variations. Additionally, four of the six models (excluding GPT-4o-mini and GPT-4o) yielded worse results in the Toddler prompt compared to the Young Child prompt.

Correctness Thresholds

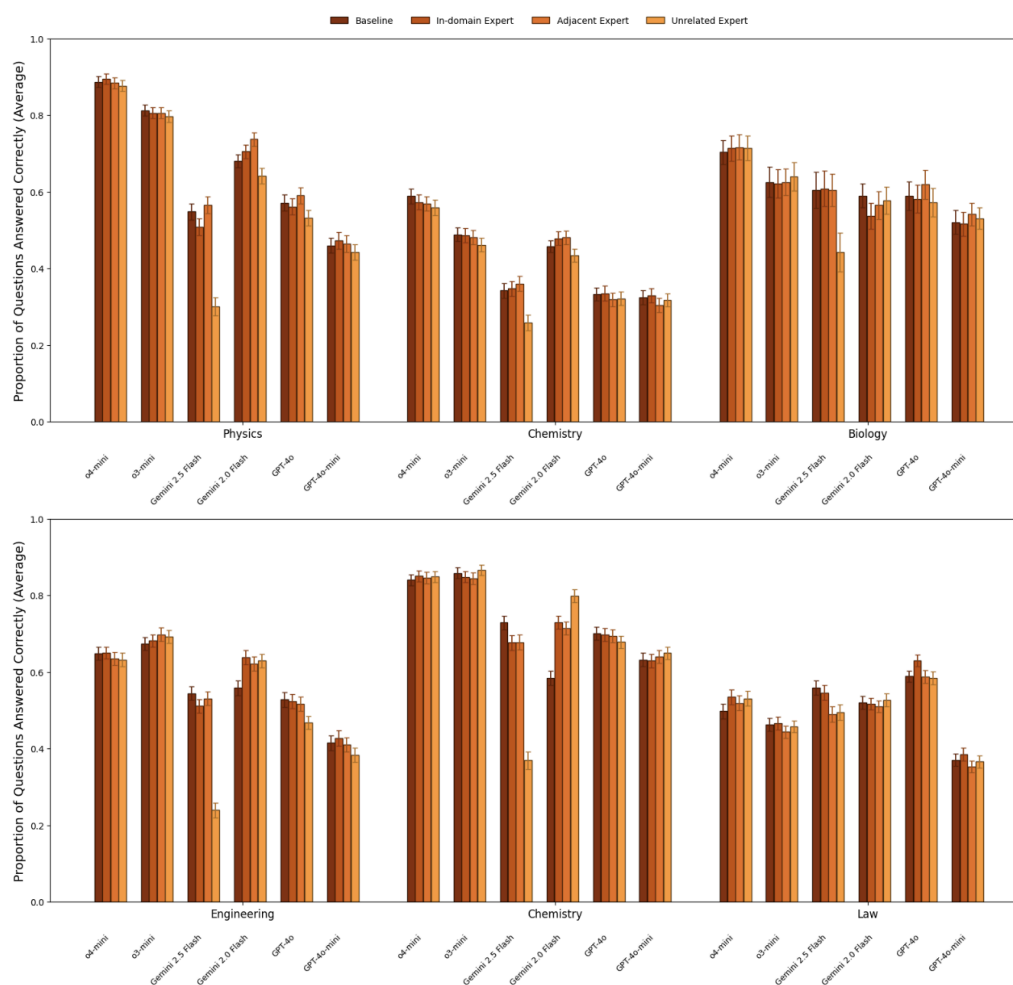
Results are overall comparable for both benchmarks when applying correctness thresholds, and in general, the effects tend to be more pronounced at the higher threshold levels. We report results for the 100%, 90% and 51% thresholds in Figures S1, S2, and S3.

Domain-Specific Persona Analysis

We next asked whether aligning expert personas with the question domain improves performance (see Figure 2). For each dataset, we compared three persona types: in-domain experts, domain-adjacent experts, and domain-unrelated experts (see How We Prompted the AI and Table S4 for definitions and mappings).

We found that domain-tailored personas do not generally improve performance. Using the GPQA Diamond dataset, there were no positive significant differences between the baseline and any of the three prompting variations (see Table S5). Using the MMLU-Pro domain subset, there were significant improvements for two models. Specifically, Gemini 2.0 Flash improved in the Engineering and Chemistry domains for all three variations and GPT-4o improved in the Law domain when comparing baseline to the in-domain expert (see Table S6). The Gemini 2.0 Flash improvements may be outliers, as another model from the same family, Gemini 2.5 Flash, exhibited no significant positive differences.

Figure 2. GPQA Diamond (top) and MMLU-Pro subset (bottom) performance across multiple domain-related prompting variations, categorized by domain



Discussion and Conclusion

Our findings indicate that, for difficult factual questions of the kind captured by GPQA Diamond and MMLU-Pro, assigning an expert persona is not a reliable way to improve accuracy. Across six models and two benchmarks, persona prompts leave performance unchanged relative to a no-persona baseline in most cases. When persona prompts do matter, they are more likely to reduce accuracy than to increase it.

The main exception is Gemini 2.0 Flash on MMLU-Pro, which shows improvements for all five expert personas overall and, at the domain level, in Engineering and Chemistry; GPT-4o also shows a small in-domain gain in Law. In contrast, low-knowledge personas (especially “Toddler”) often reduce accuracy, with “Layperson” effects varying by model (harmful in several cases, but not all). Overall, tailoring personas to match the question domain shows no consistent benefit

across models, with the limited improvements above appearing model and question-specific rather than generalizable.

A notable failure mode emerges from the domain-mismatched expert conditions in the Gemini Flash family. When instructed with an out-of-domain expert persona, Gemini 2.5 Flash frequently declines to answer questions at all. Under the Unrelated Expert condition on GPQA Diamond, the model refuses to answer an average of 10.56 out of the 25 trials per question, typically asserting that it lacks the relevant expertise or “cannot, in good conscience” select an answer. Gemini 2.0 Flash exhibits similar, but less extreme, behavior in some Physics and Chemistry conditions. This refusal-driven behavior substantially depresses measured accuracy and illustrates a broader risk: role instructions that are too narrow can cause models to under-utilize their actual knowledge.

It is important to note that these findings focus on accuracy and personas may serve additional purposes. For example, personas may change what factors the AI focuses on and how it approaches reasoning about a problem. Personas may shift what the AI prioritizes (emphasizing regulatory concerns for compliance officers vs. market opportunities for business developers) and how it expresses outcomes (confident expert vs. cautious advisor). Personas certainly play a role in shaping the tone and presentation of output (OpenAI 2025a). Beyond this, personas may serve a dual function as a prompting aid for users. Naming a specific role or perspective may help cue the user and help them better articulate and frame questions.

This study has several limitations, including testing only a subset of available models, focusing on academic benchmarks that may not reflect all real-world use cases, and examining a specific set of personas across a limited number of domains within those academic benchmarks. The consistent null results across these models and two benchmarks suggest that the strategies of assigning a persona when prompting models may be ineffective. When working on specific problems, testing multiple prompt variations may still be worthwhile given the question-level variability we have observed in previous work, but practitioners should be prepared for unpredictable results and should not expect prompting variations to provide consistent benefits. In practice, this suggests that organizations may get more value from iterating on task-specific instructions, examples, or evaluation workflows than from simply adding expert personas to prompts.

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References

- Anthropic (2024) Prompt engineering for business performance. Retrieved (October 31, 2025), <https://www.anthropic.com/news/prompt-engineering-for-business-performance>
- Anthropic (2025) Claude Developer Platform. Retrieved (November 18, 2025), <https://docs.claude.com/en/release-notes/overview>.
- Google Cloud (2024) Vertex AI Prompt Design Strategies. Retrieved (October 31, 2025), <https://cloud.google.com/vertex-ai/generative-ai/docs/learn/prompts/prompt-design-strategies>
- Google (2025) Experiment with parameter values | Generative AI on Vertex AI. Retrieved (November 18, 2025), <https://docs.cloud.google.com/vertex-ai/generative-ai/docs/learn/prompts/adjust-parameter-values>
- Kong A, Zhao S, Chen H, Li Q, Qin Y, Sun R, Zhou X, Wang E, Dong X (2024) Better Zero-Shot Reasoning with Role-Play Prompting. (March 14) <http://arxiv.org/abs/2308.07702>.
- Meincke L, Mollick ER, Mollick L, Shapiro D (2025a) Prompting Science Report 1: Prompt Engineering is Complicated and Contingent. (March 4) <https://papers.ssrn.com/abstract=5165270>.
- Meincke L, Mollick E, Mollick L, Shapiro D (2025b) Prompting Science Report 2: The Decreasing Value of Chain of Thought in Prompting. (June 8) <https://papers.ssrn.com/abstract=5285532>.
- Meincke L, Mollick E, Mollick L, Shapiro D (2025c) Prompting Science Report 3: I'll pay you or I'll kill you — but will you care? (August 1). https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5375404
- Miller E (2024) Adding Error Bars to Evals: A Statistical Approach to Language Model Evaluations. (November 1) <http://arxiv.org/abs/2411.00640>.
- OpenAI (2025a) GPT-5.1 Instant and GPT-5.1 Thinking System Card Addendum. Retrieved (December 4, 2025), https://cdn.openai.com/pdf/4173ec8d-1229-47db-96de-06d87147e07e/5_1_system_card.pdf

OpenAI (2025b) OpenAI Platform. Retrieved (November 18, 2025),
<https://platform.openai.com>

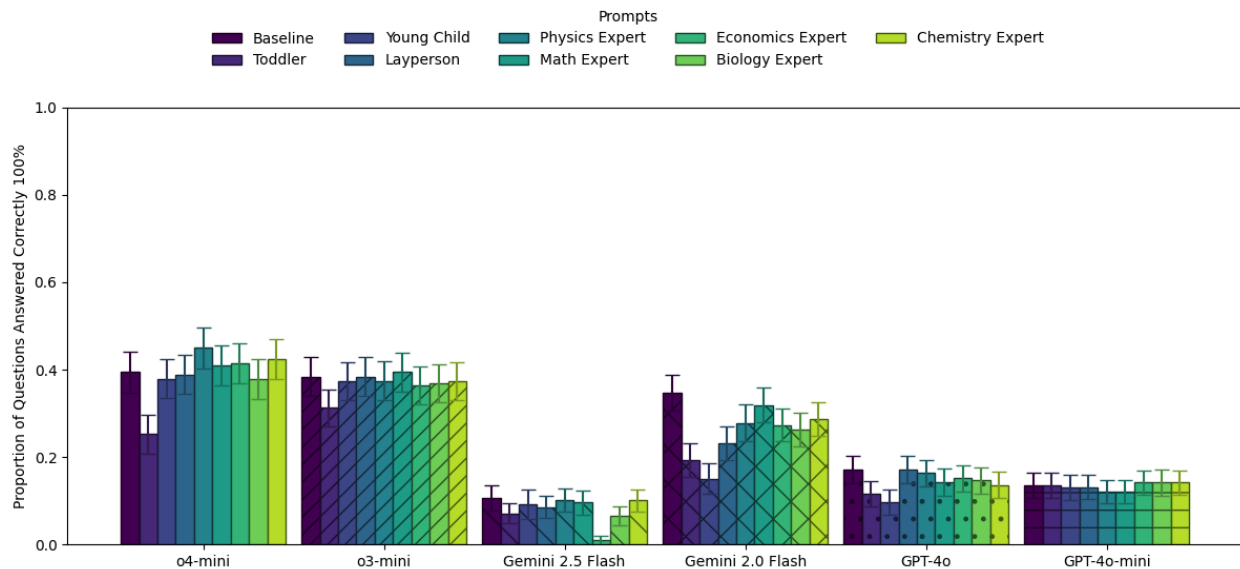
Rein D, Hou BL, Stickland AC, Petty J, Pang RY, Dirani J, Michael J, Bowman SR (2024) GPQA: A Graduate-Level Google-Proof Q&A Benchmark. *First Conference on Language Modeling*.

Sanders T (2022) Unit test writing using a multi-step prompt | OpenAI Cookbook. *Unit test writing using a multi-step prompt*. Retrieved (October 31, 2025),
https://cookbook.openai.com/examples/unit_test_writing_using_a_multi-step_prompt

Wang X, Wei J, Schuurmans D, Le Q, Chi E, Narang S, Chowdhery A, Zhou D (2023) Self-Consistency Improves Chain of Thought Reasoning in Language Models. (March 7) <http://arxiv.org/abs/2203.11171>.

Wang Y, Ma X, Zhang G, Ni Y, Chandra A, Guo S, Ren W, et al. (2024) MMLU-Pro: A More Robust and Challenging Multi-Task Language Understanding Benchmark. (November 6) <http://arxiv.org/abs/2406.01574>.

Zheng M, Pei J, Logeswaran L, Lee M, Jurgens D (2024) When “A Helpful Assistant” Is Not Really Helpful: Personas in System Prompts Do Not Improve Performances of Large Language Models. (October 9) <http://arxiv.org/abs/2311.10054>.



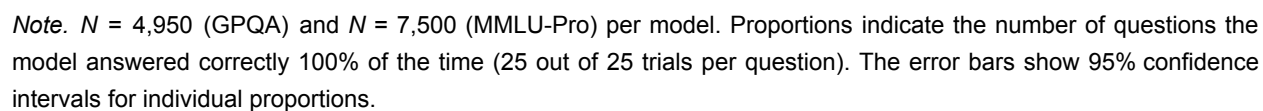
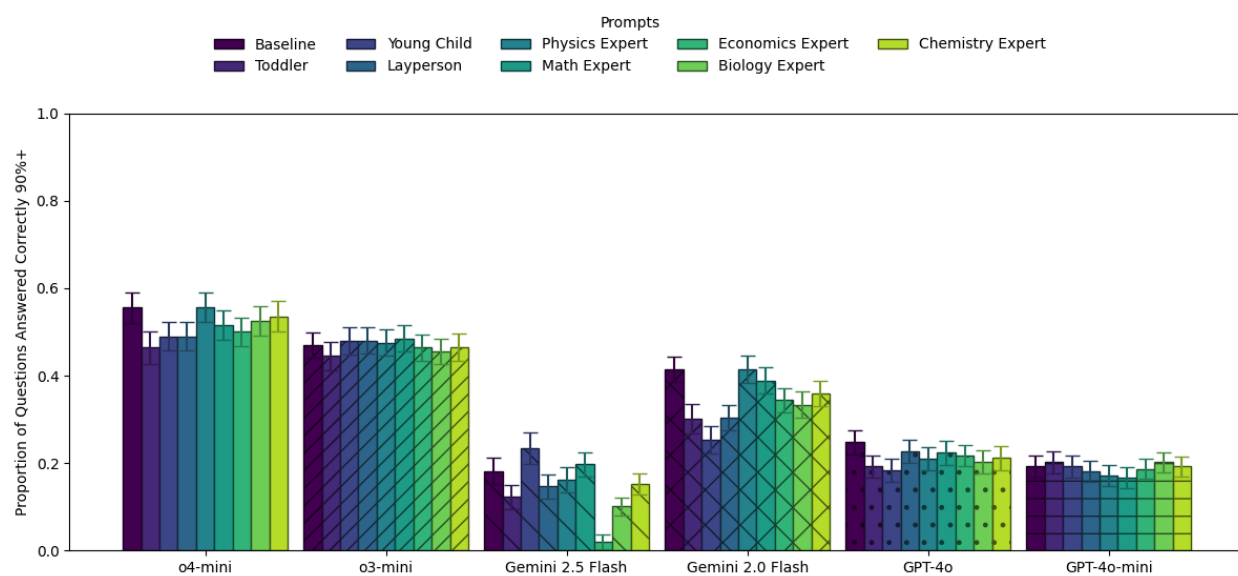


Figure S2. GPQA Diamond (top) and MMLU-Pro (bottom) subset performance across multiple prompts at 90% correctness threshold.



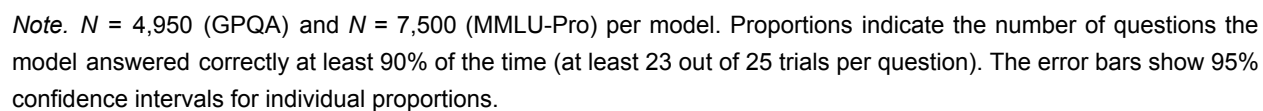
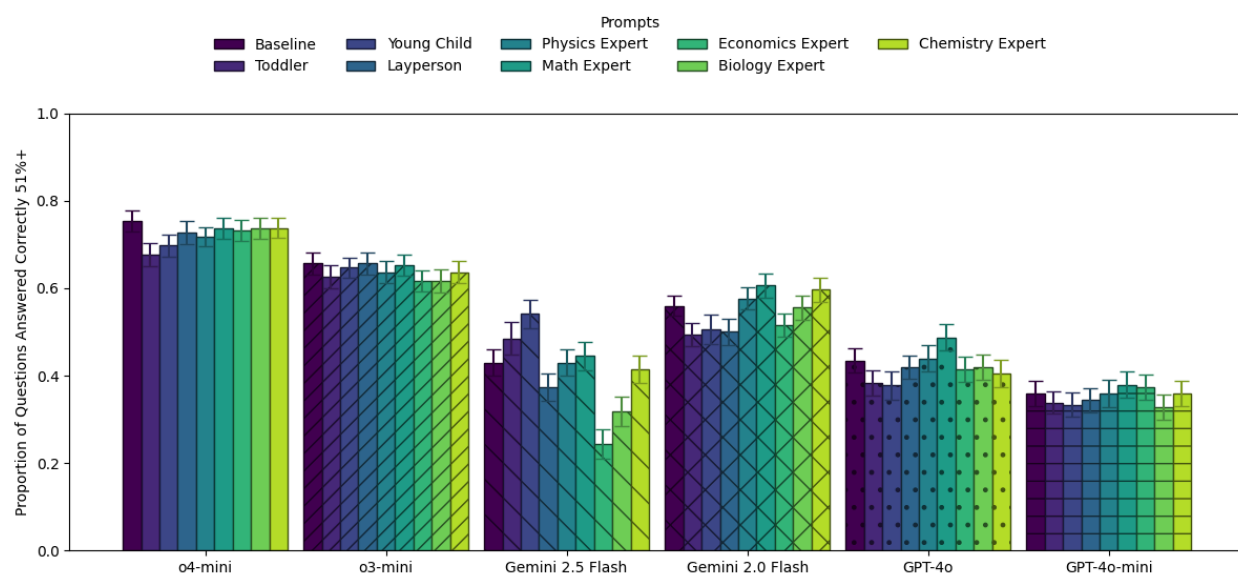


Figure S3. GPQA Diamond (top) and MMLU-Pro (bottom) subset performance across multiple prompts at 51% correctness threshold.



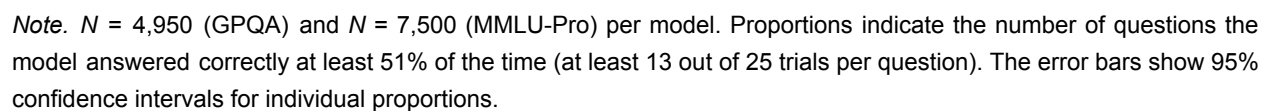
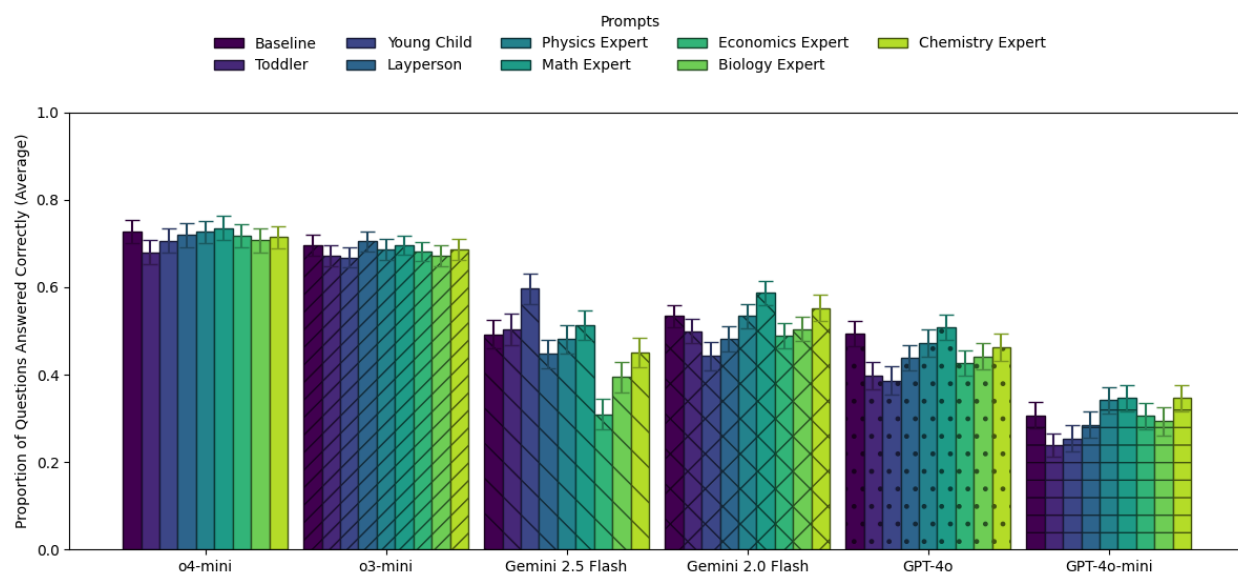
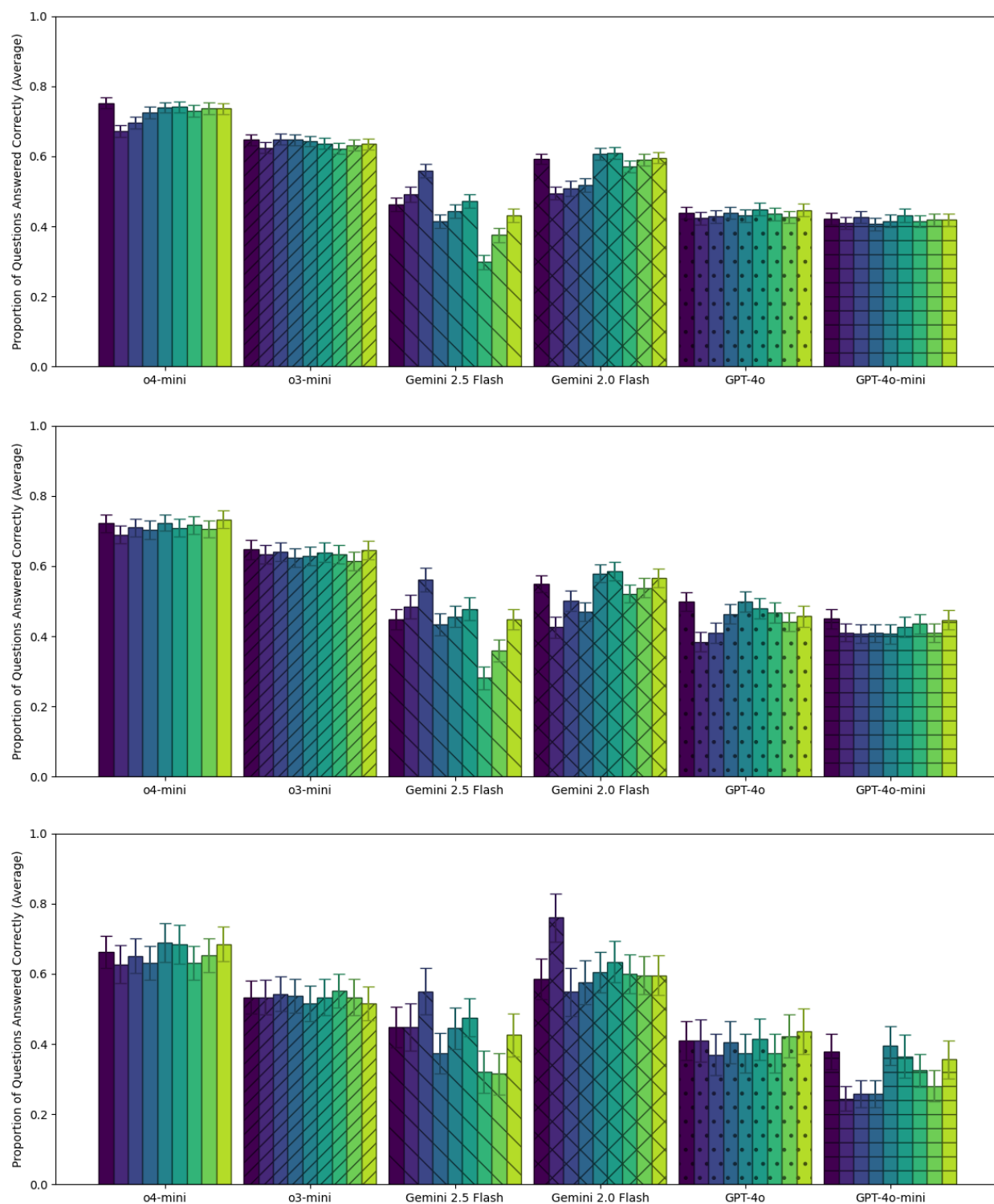


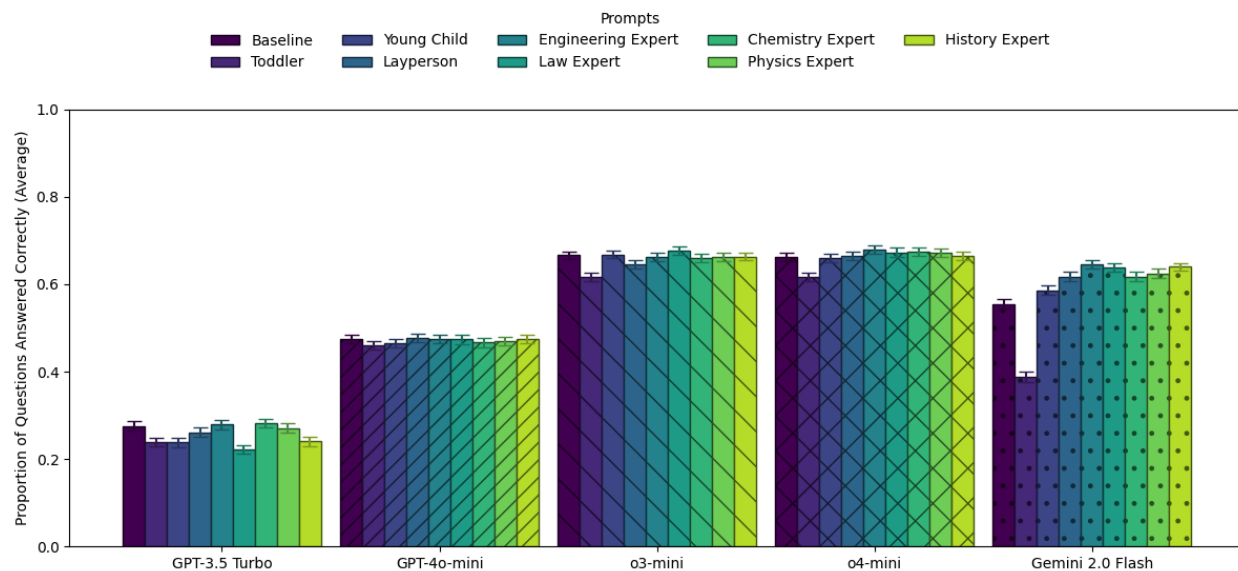
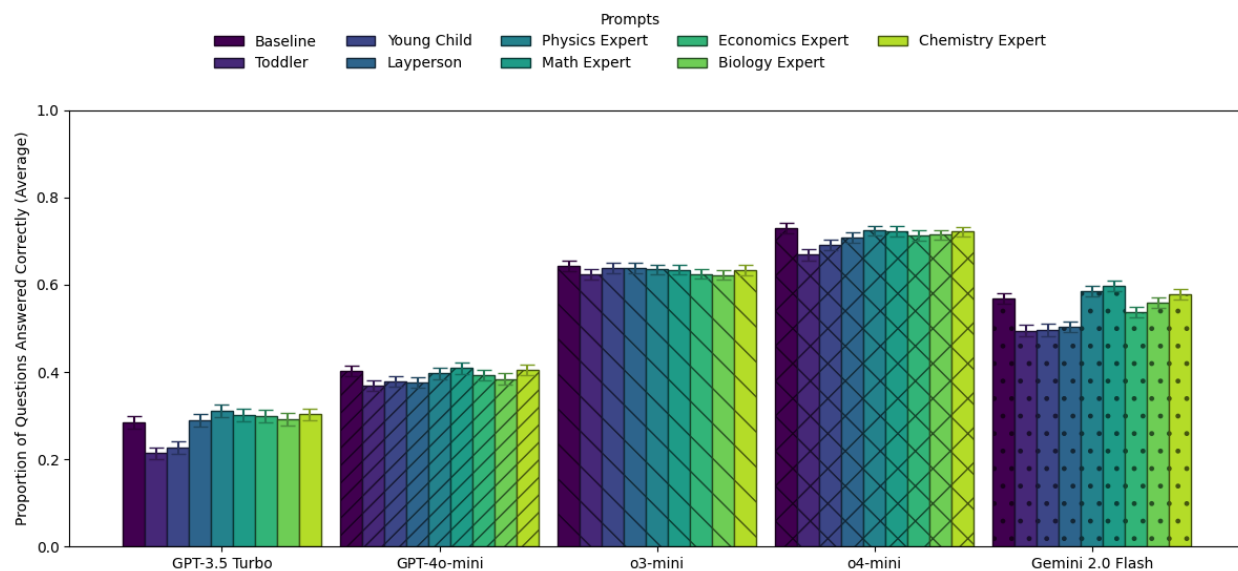
Figure S4. GPQA Diamond subset performance across multiple different prompts, on varying levels of difficulty (increasing difficulty top to bottom: “Easy Undergraduate Level”, “Hard Undergraduate Level”, “Hard Graduate Level”, “Post-graduate Level”).





Note. $N = 1,000$ for “Easy Undergraduate Level”, 2,525 for “Hard Undergraduate Level”, 1,075 for “Hard Graduate Level”, 250 for “Post-graduate Level”. The error bars show 95% confidence intervals for individual proportions.

Figure S5. GPQA Diamond (top) and MMLU-Pro (bottom) subset performance across multiple prompts, with comparison to GPT-3.5 Turbo (left-most model).



Tables

Table S1. Example questions for GPQA and MMLU-Pro.

GPQA	MMLU-Pro
What is the correct answer to this question: If a sperm from species A is injected into an egg from species B and both species have the same number of chromosomes, what would be the main cause of the resulting zygote mortality? A) Species specific zona pellucida proteins on the egg cannot bind sperms from a	A refracting telescope consists of two converging lenses separated by 100 cm. The eye-piece lens has a focal length of 20 cm. The angular magnification of the telescope is A) 10 B) 40 C) 6 D) 25

different species. B) Epistatic interactions between the genes of different species C) Chromosomal incompatibilities will cause failure of meiosis leading to death of zygote. D) Chromosomal recombination will not occur in different species. Format your response as follows: "The correct answer is (insert answer here)"	E) 15 F) 50 G) 30 H) 4 I) 5 J) 20
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Note. The example MMLU-Pro question is not part of the tested questions but from another category (physics).

Table S2. Pairwise mean comparison results between models and prompt variants (GPQA Diamond).

Model	Conditions	RD [95% CI]	Statistics
GPT-4o	Baseline - Toddler	-0.052 [-0.087, -0.016]	p = 0.004
GPT-4o	Baseline - Young Child	-0.047 [-0.120, 0.023]	p = 0.200
GPT-4o	Baseline - Layperson	-0.018 [-0.046, 0.010]	p = 0.201
GPT-4o	Baseline - Physics Expert	-0.010 [-0.082, 0.062]	p = 0.779
GPT-4o	Baseline - Math Expert	0.002 [-0.075, 0.077]	p = 0.956
GPT-4o	Baseline - Economics Expert	-0.024 [-0.051, 0.005]	p = 0.098
GPT-4o	Baseline - Biology Expert	-0.029 [-0.060, 0.003]	p = 0.064
GPT-4o	Baseline - Chemistry Expert	-0.010 [-0.036, 0.017]	p = 0.469
GPT-4o	Toddler - Young Child	0.003 [-0.067, 0.074]	p = 0.937
GPT-4o	Toddler - Layperson	0.034 [0.003, 0.063]	p = 0.028
GPT-4o	Toddler - Physics Expert	0.040 [-0.036, 0.111]	p = 0.282
GPT-4o	Toddler - Math Expert	0.054 [-0.018, 0.128]	p = 0.142
GPT-4o	Toddler - Economics Expert	0.028 [0.001, 0.056]	p = 0.045
GPT-4o	Toddler - Biology Expert	0.023 [-0.004, 0.048]	p = 0.082
GPT-4o	Toddler - Chemistry Expert	0.042 [0.009, 0.076]	p = 0.016
GPT-4o	Young Child - Layperson	0.028 [-0.042, 0.099]	p = 0.444
GPT-4o	Young Child - Physics Expert	0.037 [0.002, 0.073]	p = 0.040
GPT-4o	Young Child - Math Expert	0.052 [-0.023, 0.130]	p = 0.185
GPT-4o	Young Child - Economics Expert	0.025 [-0.046, 0.092]	p = 0.490
GPT-4o	Young Child - Biology Expert	0.019 [-0.051, 0.089]	p = 0.591
GPT-4o	Young Child - Chemistry Expert	0.036 [-0.028, 0.105]	p = 0.285
GPT-4o	Layperson - Physics Expert	0.009 [-0.068, 0.083]	p = 0.827

GPT-4o	Layperson - Math Expert	0.021 [-0.054, 0.096]	p = 0.581
GPT-4o	Layperson - Economics Expert	-0.005 [-0.030, 0.019]	p = 0.684
GPT-4o	Layperson - Biology Expert	-0.011 [-0.035, 0.014]	p = 0.393
GPT-4o	Layperson - Chemistry Expert	0.008 [-0.021, 0.038]	p = 0.569
GPT-4o	Physics Expert - Math Expert	0.015 [-0.059, 0.092]	p = 0.704
GPT-4o	Physics Expert - Economics Expert	-0.013 [-0.087, 0.061]	p = 0.739
GPT-4o	Physics Expert - Biology Expert	-0.018 [-0.090, 0.056]	p = 0.639
GPT-4o	Physics Expert - Chemistry Expert	-0.001 [-0.071, 0.070]	p = 0.983
GPT-4o	Math Expert - Economics Expert	-0.027 [-0.099, 0.046]	p = 0.490
GPT-4o	Math Expert - Biology Expert	-0.032 [-0.104, 0.045]	p = 0.412
GPT-4o	Math Expert - Chemistry Expert	-0.013 [-0.090, 0.061]	p = 0.721
GPT-4o	Economics Expert - Biology Expert	-0.005 [-0.027, 0.017]	p = 0.655
GPT-4o	Economics Expert - Chemistry Expert	0.014 [-0.013, 0.040]	p = 0.317
GPT-4o	Biology Expert - Chemistry Expert	0.019 [-0.009, 0.048]	p = 0.187
GPT-4o-mini	Baseline - Toddler	-0.034 [-0.070, 0.003]	p = 0.075
GPT-4o-mini	Baseline - Young Child	-0.023 [-0.051, 0.004]	p = 0.100
GPT-4o-mini	Baseline - Layperson	-0.026 [-0.052, 0.000]	p = 0.052
GPT-4o-mini	Baseline - Physics Expert	-0.005 [-0.031, 0.020]	p = 0.687
GPT-4o-mini	Baseline - Math Expert	0.006 [-0.020, 0.033]	p = 0.641
GPT-4o-mini	Baseline - Economics Expert	-0.010 [-0.034, 0.013]	p = 0.405
GPT-4o-mini	Baseline - Biology Expert	-0.018 [-0.045, 0.009]	p = 0.194
GPT-4o-mini	Baseline - Chemistry Expert	0.003 [-0.024, 0.030]	p = 0.844
GPT-4o-mini	Toddler - Young Child	0.011 [-0.016, 0.038]	p = 0.452
GPT-4o-mini	Toddler - Layperson	0.007 [-0.021, 0.038]	p = 0.635
GPT-4o-mini	Toddler - Physics Expert	0.028 [-0.009, 0.067]	p = 0.139
GPT-4o-mini	Toddler - Math Expert	0.040 [0.002, 0.077]	p = 0.037
GPT-4o-mini	Toddler - Economics Expert	0.024 [-0.008, 0.054]	p = 0.142
GPT-4o-mini	Toddler - Biology Expert	0.015 [-0.013, 0.045]	p = 0.299
GPT-4o-mini	Toddler - Chemistry Expert	0.036 [-0.001, 0.074]	p = 0.062
GPT-4o-mini	Young Child - Layperson	-0.003 [-0.026, 0.019]	p = 0.784
GPT-4o-mini	Young Child - Physics Expert	0.018 [-0.013, 0.049]	p = 0.237
GPT-4o-mini	Young Child - Math Expert	0.029 [-0.001, 0.061]	p = 0.061
GPT-4o-mini	Young Child - Economics Expert	0.013 [-0.010, 0.037]	p = 0.281
GPT-4o-mini	Young Child - Biology Expert	0.005 [-0.020, 0.029]	p = 0.708
GPT-4o-mini	Young Child - Chemistry Expert	0.026 [-0.005, 0.057]	p = 0.095

GPT-4o-mini	Layperson - Physics Expert	0.021 [-0.006, 0.049]	p = 0.144
GPT-4o-mini	Layperson - Math Expert	0.033 [0.003, 0.063]	p = 0.031
GPT-4o-mini	Layperson - Economics Expert	0.016 [-0.006, 0.039]	p = 0.163
GPT-4o-mini	Layperson - Biology Expert	0.008 [-0.016, 0.033]	p = 0.514
GPT-4o-mini	Layperson - Chemistry Expert	0.029 [-0.000, 0.058]	p = 0.050
GPT-4o-mini	Physics Expert - Math Expert	0.011 [-0.013, 0.036]	p = 0.357
GPT-4o-mini	Physics Expert - Economics Expert	-0.005 [-0.030, 0.021]	p = 0.709
GPT-4o-mini	Physics Expert - Biology Expert	-0.013 [-0.042, 0.016]	p = 0.380
GPT-4o-mini	Physics Expert - Chemistry Expert	0.008 [-0.019, 0.035]	p = 0.570
GPT-4o-mini	Math Expert - Economics Expert	-0.016 [-0.044, 0.011]	p = 0.249
GPT-4o-mini	Math Expert - Biology Expert	-0.025 [-0.053, 0.004]	p = 0.098
GPT-4o-mini	Math Expert - Chemistry Expert	-0.003 [-0.030, 0.024]	p = 0.800
GPT-4o-mini	Economics Expert - Biology Expert	-0.008 [-0.032, 0.016]	p = 0.503
GPT-4o-mini	Economics Expert - Chemistry Expert	0.013 [-0.016, 0.039]	p = 0.343
GPT-4o-mini	Biology Expert - Chemistry Expert	0.021 [-0.007, 0.050]	p = 0.147
o3-mini	Baseline - Toddler	-0.019 [-0.040, 0.001]	p = 0.073
o3-mini	Baseline - Young Child	-0.005 [-0.025, 0.015]	p = 0.652
o3-mini	Baseline - Layperson	-0.004 [-0.025, 0.016]	p = 0.700
o3-mini	Baseline - Physics Expert	-0.008 [-0.028, 0.012]	p = 0.420
o3-mini	Baseline - Math Expert	-0.008 [-0.027, 0.011]	p = 0.399
o3-mini	Baseline - Economics Expert	-0.018 [-0.039, 0.002]	p = 0.082
o3-mini	Baseline - Biology Expert	-0.020 [-0.041, -0.001]	p = 0.047
o3-mini	Baseline - Chemistry Expert	-0.009 [-0.029, 0.011]	p = 0.373
o3-mini	Toddler - Young Child	0.014 [-0.005, 0.034]	p = 0.158
o3-mini	Toddler - Layperson	0.015 [-0.005, 0.034]	p = 0.136
o3-mini	Toddler - Physics Expert	0.011 [-0.009, 0.031]	p = 0.272
o3-mini	Toddler - Math Expert	0.011 [-0.010, 0.030]	p = 0.274
o3-mini	Toddler - Economics Expert	0.001 [-0.019, 0.020]	p = 0.928
o3-mini	Toddler - Biology Expert	-0.001 [-0.022, 0.019]	p = 0.892
o3-mini	Toddler - Chemistry Expert	0.010 [-0.011, 0.030]	p = 0.349
o3-mini	Young Child - Layperson	0.001 [-0.019, 0.021]	p = 0.964
o3-mini	Young Child - Physics Expert	-0.003 [-0.023, 0.016]	p = 0.739
o3-mini	Young Child - Math Expert	-0.003 [-0.022, 0.016]	p = 0.727
o3-mini	Young Child - Economics Expert	-0.014 [-0.034, 0.007]	p = 0.207
o3-mini	Young Child - Biology Expert	-0.016 [-0.036, 0.004]	p = 0.119

o3-mini	Young Child - Chemistry Expert	-0.004 [-0.025, 0.016]	p = 0.675
o3-mini	Layperson - Physics Expert	-0.004 [-0.023, 0.016]	p = 0.705
o3-mini	Layperson - Math Expert	-0.004 [-0.024, 0.016]	p = 0.688
o3-mini	Layperson - Economics Expert	-0.014 [-0.034, 0.005]	p = 0.162
o3-mini	Layperson - Biology Expert	-0.016 [-0.037, 0.004]	p = 0.120
o3-mini	Layperson - Chemistry Expert	-0.005 [-0.027, 0.017]	p = 0.664
o3-mini	Physics Expert - Math Expert	-0.000 [-0.019, 0.019]	p = 0.985
o3-mini	Physics Expert - Economics Expert	-0.010 [-0.030, 0.010]	p = 0.311
o3-mini	Physics Expert - Biology Expert	-0.012 [-0.032, 0.007]	p = 0.198
o3-mini	Physics Expert - Chemistry Expert	-0.001 [-0.020, 0.018]	p = 0.915
o3-mini	Math Expert - Economics Expert	-0.010 [-0.029, 0.009]	p = 0.294
o3-mini	Math Expert - Biology Expert	-0.012 [-0.031, 0.007]	p = 0.202
o3-mini	Math Expert - Chemistry Expert	-0.001 [-0.020, 0.018]	p = 0.931
o3-mini	Economics Expert - Biology Expert	-0.002 [-0.021, 0.017]	p = 0.817
o3-mini	Economics Expert - Chemistry Expert	0.009 [-0.013, 0.030]	p = 0.394
o3-mini	Biology Expert - Chemistry Expert	0.011 [-0.009, 0.032]	p = 0.264
o4-mini	Baseline - Toddler	-0.061 [-0.086, -0.036]	p < 0.001
o4-mini	Baseline - Young Child	-0.037 [-0.059, -0.016]	p < 0.001
o4-mini	Baseline - Layperson	-0.022 [-0.043, -0.003]	p = 0.030
o4-mini	Baseline - Physics Expert	-0.005 [-0.024, 0.014]	p = 0.612
o4-mini	Baseline - Math Expert	-0.007 [-0.026, 0.013]	p = 0.526
o4-mini	Baseline - Economics Expert	-0.018 [-0.037, 0.001]	p = 0.075
o4-mini	Baseline - Biology Expert	-0.015 [-0.035, 0.004]	p = 0.123
o4-mini	Baseline - Chemistry Expert	-0.008 [-0.027, 0.011]	p = 0.404
o4-mini	Toddler - Young Child	0.024 [0.001, 0.045]	p = 0.036
o4-mini	Toddler - Layperson	0.039 [0.015, 0.061]	p < 0.001
o4-mini	Toddler - Physics Expert	0.056 [0.031, 0.081]	p < 0.001
o4-mini	Toddler - Math Expert	0.054 [0.029, 0.079]	p < 0.001
o4-mini	Toddler - Economics Expert	0.043 [0.020, 0.067]	p < 0.001
o4-mini	Toddler - Biology Expert	0.046 [0.021, 0.070]	p < 0.001
o4-mini	Toddler - Chemistry Expert	0.053 [0.027, 0.079]	p < 0.001
o4-mini	Young Child - Layperson	0.015 [-0.005, 0.035]	p = 0.150
o4-mini	Young Child - Physics Expert	0.032 [0.011, 0.054]	p = 0.003
o4-mini	Young Child - Math Expert	0.031 [0.010, 0.052]	p = 0.004
o4-mini	Young Child - Economics Expert	0.020 [-0.001, 0.040]	p = 0.055

o4-mini	Young Child - Biology Expert	0.022 [0.001, 0.045]	p = 0.050
o4-mini	Young Child - Chemistry Expert	0.029 [0.008, 0.051]	p = 0.008
o4-mini	Layperson - Physics Expert	0.017 [-0.004, 0.038]	p = 0.106
o4-mini	Layperson - Math Expert	0.016 [-0.005, 0.037]	p = 0.138
o4-mini	Layperson - Economics Expert	0.005 [-0.015, 0.025]	p = 0.619
o4-mini	Layperson - Biology Expert	0.007 [-0.013, 0.027]	p = 0.488
o4-mini	Layperson - Chemistry Expert	0.014 [-0.006, 0.035]	p = 0.201
o4-mini	Physics Expert - Math Expert	-0.002 [-0.020, 0.018]	p = 0.876
o4-mini	Physics Expert - Economics Expert	-0.013 [-0.033, 0.007]	p = 0.213
o4-mini	Physics Expert - Biology Expert	-0.010 [-0.031, 0.011]	p = 0.332
o4-mini	Physics Expert - Chemistry Expert	-0.003 [-0.022, 0.016]	p = 0.730
o4-mini	Math Expert - Economics Expert	-0.011 [-0.030, 0.009]	p = 0.281
o4-mini	Math Expert - Biology Expert	-0.009 [-0.029, 0.011]	p = 0.390
o4-mini	Math Expert - Chemistry Expert	-0.002 [-0.021, 0.017]	p = 0.861
o4-mini	Economics Expert - Biology Expert	0.002 [-0.019, 0.022]	p = 0.823
o4-mini	Economics Expert - Chemistry Expert	0.009 [-0.011, 0.029]	p = 0.358
o4-mini	Biology Expert - Chemistry Expert	0.007 [-0.014, 0.028]	p = 0.518
Gemini 2.0 Flash	Baseline - Toddler	-0.078 [-0.166, 0.009]	p = 0.082
Gemini 2.0 Flash	Baseline - Young Child	-0.076 [-0.166, 0.018]	p = 0.110
Gemini 2.0 Flash	Baseline - Layperson	-0.073 [-0.161, 0.014]	p = 0.103
Gemini 2.0 Flash	Baseline - Physics Expert	-0.005 [-0.090, 0.081]	p = 0.910
Gemini 2.0 Flash	Baseline - Math Expert	0.006 [-0.077, 0.091]	p = 0.887
Gemini 2.0 Flash	Baseline - Economics Expert	-0.053 [-0.138, 0.031]	p = 0.216
Gemini 2.0 Flash	Baseline - Biology Expert	-0.033 [-0.116, 0.051]	p = 0.439
Gemini 2.0 Flash	Baseline - Chemistry Expert	-0.012 [-0.098, 0.074]	p = 0.780
Gemini 2.0 Flash	Toddler - Young Child	0.001 [-0.063, 0.065]	p = 0.963
Gemini 2.0 Flash	Toddler - Layperson	0.010 [-0.055, 0.074]	p = 0.748
Gemini 2.0 Flash	Toddler - Physics Expert	0.088 [0.044, 0.133]	p < 0.001
Gemini 2.0 Flash	Toddler - Math Expert	0.100 [0.053, 0.147]	p < 0.001
Gemini 2.0 Flash	Toddler - Economics Expert	0.047 [0.003, 0.089]	p = 0.033
Gemini 2.0 Flash	Toddler - Biology Expert	0.064 [0.022, 0.105]	p = 0.003
Gemini 2.0 Flash	Toddler - Chemistry Expert	0.083 [0.038, 0.128]	p < 0.001
Gemini 2.0 Flash	Young Child - Layperson	0.009 [-0.056, 0.070]	p = 0.780
Gemini 2.0 Flash	Young Child - Physics Expert	0.087 [0.017, 0.158]	p = 0.014
Gemini 2.0 Flash	Young Child - Math Expert	0.099 [0.031, 0.168]	p = 0.004

Gemini 2.0 Flash	Young Child - Economics Expert	0.045 [-0.024, 0.114]	p = 0.199
Gemini 2.0 Flash	Young Child - Biology Expert	0.063 [-0.006, 0.131]	p = 0.075
Gemini 2.0 Flash	Young Child - Chemistry Expert	0.081 [0.013, 0.150]	p = 0.023
Gemini 2.0 Flash	Layperson - Physics Expert	0.072 [0.006, 0.138]	p = 0.035
Gemini 2.0 Flash	Layperson - Math Expert	0.084 [0.019, 0.149]	p = 0.011
Gemini 2.0 Flash	Layperson - Economics Expert	0.031 [-0.033, 0.099]	p = 0.360
Gemini 2.0 Flash	Layperson - Biology Expert	0.048 [-0.015, 0.112]	p = 0.138
Gemini 2.0 Flash	Layperson - Chemistry Expert	0.067 [0.002, 0.131]	p = 0.043
Gemini 2.0 Flash	Physics Expert - Math Expert	0.012 [-0.012, 0.037]	p = 0.350
Gemini 2.0 Flash	Physics Expert - Economics Expert	-0.047 [-0.080, -0.015]	p = 0.006
Gemini 2.0 Flash	Physics Expert - Biology Expert	-0.026 [-0.054, 0.001]	p = 0.062
Gemini 2.0 Flash	Physics Expert - Chemistry Expert	-0.007 [-0.031, 0.017]	p = 0.566
Gemini 2.0 Flash	Math Expert - Economics Expert	-0.059 [-0.093, -0.026]	p < 0.001
Gemini 2.0 Flash	Math Expert - Biology Expert	-0.038 [-0.067, -0.011]	p = 0.007
Gemini 2.0 Flash	Math Expert - Chemistry Expert	-0.019 [-0.044, 0.006]	p = 0.140
Gemini 2.0 Flash	Economics Expert - Biology Expert	0.021 [-0.004, 0.046]	p = 0.114
Gemini 2.0 Flash	Economics Expert - Chemistry Expert	0.040 [0.009, 0.070]	p = 0.007
Gemini 2.0 Flash	Biology Expert - Chemistry Expert	0.019 [-0.007, 0.045]	p = 0.147
Gemini 2.5 Flash	Baseline - Toddler	0.021 [-0.043, 0.084]	p = 0.526
Gemini 2.5 Flash	Baseline - Young Child	0.098 [0.029, 0.164]	p = 0.005
Gemini 2.5 Flash	Baseline - Layperson	-0.039 [-0.071, -0.005]	p = 0.023
Gemini 2.5 Flash	Baseline - Physics Expert	-0.012 [-0.038, 0.015]	p = 0.394
Gemini 2.5 Flash	Baseline - Math Expert	0.018 [-0.008, 0.044]	p = 0.169
Gemini 2.5 Flash	Baseline - Economics Expert	-0.163 [-0.205, -0.121]	p < 0.001
Gemini 2.5 Flash	Baseline - Biology Expert	-0.087 [-0.124, -0.049]	p < 0.001
Gemini 2.5 Flash	Baseline - Chemistry Expert	-0.027 [-0.055, 0.001]	p = 0.054
Gemini 2.5 Flash	Toddler - Young Child	0.077 [0.027, 0.127]	p = 0.002
Gemini 2.5 Flash	Toddler - Layperson	-0.059 [-0.120, 0.003]	p = 0.058
Gemini 2.5 Flash	Toddler - Physics Expert	-0.032 [-0.096, 0.030]	p = 0.311
Gemini 2.5 Flash	Toddler - Math Expert	-0.006 [-0.070, 0.057]	p = 0.866
Gemini 2.5 Flash	Toddler - Economics Expert	-0.186 [-0.248, -0.124]	p < 0.001
Gemini 2.5 Flash	Toddler - Biology Expert	-0.109 [-0.171, -0.049]	p < 0.001
Gemini 2.5 Flash	Toddler - Chemistry Expert	-0.048 [-0.110, 0.014]	p = 0.134
Gemini 2.5 Flash	Young Child - Layperson	-0.136 [-0.199, -0.071]	p < 0.001
Gemini 2.5 Flash	Young Child - Physics Expert	-0.109 [-0.173, -0.044]	p = 0.001

Gemini 2.5 Flash	Young Child - Math Expert	-0.082 [-0.147, -0.021]	p = 0.009
Gemini 2.5 Flash	Young Child - Economics Expert	-0.262 [-0.330, -0.195]	p < 0.001
Gemini 2.5 Flash	Young Child - Biology Expert	-0.186 [-0.255, -0.119]	p < 0.001
Gemini 2.5 Flash	Young Child - Chemistry Expert	-0.125 [-0.190, -0.060]	p < 0.001
Gemini 2.5 Flash	Layperson - Physics Expert	0.027 [-0.004, 0.057]	p = 0.085
Gemini 2.5 Flash	Layperson - Math Expert	0.057 [0.021, 0.093]	p = 0.002
Gemini 2.5 Flash	Layperson - Economics Expert	-0.124 [-0.161, -0.086]	p < 0.001
Gemini 2.5 Flash	Layperson - Biology Expert	-0.049 [-0.080, -0.016]	p = 0.003
Gemini 2.5 Flash	Layperson - Chemistry Expert	0.011 [-0.019, 0.041]	p = 0.464
Gemini 2.5 Flash	Physics Expert - Math Expert	0.030 [0.002, 0.060]	p = 0.041
Gemini 2.5 Flash	Physics Expert - Economics Expert	-0.151 [-0.192, -0.111]	p < 0.001
Gemini 2.5 Flash	Physics Expert - Biology Expert	-0.076 [-0.111, -0.041]	p < 0.001
Gemini 2.5 Flash	Physics Expert - Chemistry Expert	-0.016 [-0.040, 0.010]	p = 0.226
Gemini 2.5 Flash	Math Expert - Economics Expert	-0.181 [-0.225, -0.137]	p < 0.001
Gemini 2.5 Flash	Math Expert - Biology Expert	-0.106 [-0.146, -0.065]	p < 0.001
Gemini 2.5 Flash	Math Expert - Chemistry Expert	-0.046 [-0.077, -0.015]	p = 0.004
Gemini 2.5 Flash	Economics Expert - Biology Expert	0.075 [0.044, 0.106]	p < 0.001
Gemini 2.5 Flash	Economics Expert - Chemistry Expert	0.135 [0.098, 0.174]	p < 0.001
Gemini 2.5 Flash	Biology Expert - Chemistry Expert	0.060 [0.027, 0.093]	p < 0.001

Note. $N = 198$ questions in each condition with 25 trials. Paired bootstrap-permutation tests (5,000 replicates). P-values represent the proportion of permuted differences with absolute values exceeding observed differences under the null hypothesis. All p-values are uncorrected for multiple comparisons.

Table S3. Pairwise mean comparison results between models and prompt variants (MMLU-Pro).

Model	Conditions	RD [95% CI]	Statistics
GPT-4o	Baseline - Toddler	-0.040 [-0.092, 0.013]	p = 0.134
GPT-4o	Baseline - Young Child	-0.056 [-0.115, 0.002]	p = 0.060
GPT-4o	Baseline - Layperson	-0.015 [-0.068, 0.038]	p = 0.584
GPT-4o	Baseline - Engineering Expert	-0.003 [-0.054, 0.047]	p = 0.893
GPT-4o	Baseline - Law Expert	-0.013 [-0.032, 0.006]	p = 0.175
GPT-4o	Baseline - Chemistry Expert	-0.022 [-0.073, 0.027]	p = 0.395
GPT-4o	Baseline - Physics Expert	-0.009 [-0.067, 0.050]	p = 0.773
GPT-4o	Baseline - History Expert	-0.011 [-0.063, 0.043]	p = 0.678
GPT-4o	Toddler - Young Child	-0.016 [-0.072, 0.041]	p = 0.602

GPT-4o	Toddler - Layperson	0.025 [0.006, 0.045]	p = 0.013
GPT-4o	Toddler - Engineering Expert	0.037 [0.016, 0.058]	p < 0.001
GPT-4o	Toddler - Law Expert	0.027 [-0.027, 0.081]	p = 0.332
GPT-4o	Toddler - Chemistry Expert	0.018 [-0.003, 0.038]	p = 0.096
GPT-4o	Toddler - Physics Expert	0.031 [-0.024, 0.088]	p = 0.283
GPT-4o	Toddler - History Expert	0.029 [0.008, 0.050]	p = 0.008
GPT-4o	Young Child - Layperson	0.041 [-0.017, 0.102]	p = 0.171
GPT-4o	Young Child - Engineering Expert	0.053 [-0.005, 0.111]	p = 0.078
GPT-4o	Young Child - Law Expert	0.043 [-0.014, 0.102]	p = 0.145
GPT-4o	Young Child - Chemistry Expert	0.034 [-0.026, 0.092]	p = 0.264
GPT-4o	Young Child - Physics Expert	0.048 [-0.006, 0.102]	p = 0.080
GPT-4o	Young Child - History Expert	0.045 [-0.013, 0.102]	p = 0.123
GPT-4o	Layperson - Engineering Expert	0.011 [-0.007, 0.029]	p = 0.218
GPT-4o	Layperson - Law Expert	0.002 [-0.053, 0.055]	p = 0.944
GPT-4o	Layperson - Chemistry Expert	-0.008 [-0.027, 0.011]	p = 0.426
GPT-4o	Layperson - Physics Expert	0.006 [-0.054, 0.065]	p = 0.838
GPT-4o	Layperson - History Expert	0.003 [-0.014, 0.021]	p = 0.704
GPT-4o	Engineering Expert - Law Expert	-0.009 [-0.063, 0.043]	p = 0.721
GPT-4o	Engineering Expert - Chemistry Expert	-0.019 [-0.037, -0.002]	p = 0.037
GPT-4o	Engineering Expert - Physics Expert	-0.005 [-0.064, 0.052]	p = 0.851
GPT-4o	Engineering Expert - History Expert	-0.008 [-0.025, 0.009]	p = 0.381
GPT-4o	Law Expert - Chemistry Expert	-0.010 [-0.063, 0.042]	p = 0.723
GPT-4o	Law Expert - Physics Expert	0.004 [-0.054, 0.063]	p = 0.895
GPT-4o	Law Expert - History Expert	0.002 [-0.052, 0.056]	p = 0.950
GPT-4o	Chemistry Expert - Physics Expert	0.013 [-0.044, 0.072]	p = 0.655
GPT-4o	Chemistry Expert - History Expert	0.011 [-0.007, 0.029]	p = 0.219
GPT-4o	Physics Expert - History Expert	-0.002 [-0.061, 0.057]	p = 0.933
GPT-4o-mini	Baseline - Toddler	-0.014 [-0.033, 0.005]	p = 0.170
GPT-4o-mini	Baseline - Young Child	-0.008 [-0.026, 0.009]	p = 0.355
GPT-4o-mini	Baseline - Layperson	0.003 [-0.014, 0.021]	p = 0.706
GPT-4o-mini	Baseline - Engineering Expert	0.003 [-0.052, 0.057]	p = 0.922
GPT-4o-mini	Baseline - Law Expert	-0.000 [-0.019, 0.019]	p = 0.966
GPT-4o-mini	Baseline - Chemistry Expert	-0.007 [-0.025, 0.012]	p = 0.435
GPT-4o-mini	Baseline - Physics Expert	-0.005 [-0.022, 0.013]	p = 0.588
GPT-4o-mini	Baseline - History Expert	0.001 [-0.018, 0.020]	p = 0.898

GPT-4o-mini	Toddler - Young Child	0.005 [-0.012, 0.023]	p = 0.557
GPT-4o-mini	Toddler - Layperson	0.017 [-0.003, 0.037]	p = 0.091
GPT-4o-mini	Toddler - Engineering Expert	0.015 [-0.039, 0.068]	p = 0.592
GPT-4o-mini	Toddler - Law Expert	0.013 [-0.008, 0.033]	p = 0.201
GPT-4o-mini	Toddler - Chemistry Expert	0.007 [-0.012, 0.026]	p = 0.503
GPT-4o-mini	Toddler - Physics Expert	0.009 [-0.010, 0.028]	p = 0.364
GPT-4o-mini	Toddler - History Expert	0.015 [-0.005, 0.034]	p = 0.134
GPT-4o-mini	Young Child - Layperson	0.012 [-0.006, 0.030]	p = 0.202
GPT-4o-mini	Young Child - Engineering Expert	0.011 [-0.043, 0.066]	p = 0.703
GPT-4o-mini	Young Child - Law Expert	0.008 [-0.011, 0.027]	p = 0.407
GPT-4o-mini	Young Child - Chemistry Expert	0.001 [-0.017, 0.019]	p = 0.905
GPT-4o-mini	Young Child - Physics Expert	0.003 [-0.015, 0.021]	p = 0.709
GPT-4o-mini	Young Child - History Expert	0.010 [-0.009, 0.028]	p = 0.310
GPT-4o-mini	Layperson - Engineering Expert	-0.001 [-0.055, 0.054]	p = 0.978
GPT-4o-mini	Layperson - Law Expert	-0.004 [-0.023, 0.015]	p = 0.701
GPT-4o-mini	Layperson - Chemistry Expert	-0.011 [-0.030, 0.008]	p = 0.268
GPT-4o-mini	Layperson - Physics Expert	-0.008 [-0.027, 0.010]	p = 0.367
GPT-4o-mini	Layperson - History Expert	-0.002 [-0.020, 0.015]	p = 0.817
GPT-4o-mini	Engineering Expert - Law Expert	-0.003 [-0.058, 0.051]	p = 0.912
GPT-4o-mini	Engineering Expert - Chemistry Expert	-0.010 [-0.064, 0.044]	p = 0.715
GPT-4o-mini	Engineering Expert - Physics Expert	-0.008 [-0.060, 0.046]	p = 0.784
GPT-4o-mini	Engineering Expert - History Expert	-0.001 [-0.055, 0.051]	p = 0.973
GPT-4o-mini	Law Expert - Chemistry Expert	-0.007 [-0.027, 0.013]	p = 0.491
GPT-4o-mini	Law Expert - Physics Expert	-0.005 [-0.023, 0.014]	p = 0.631
GPT-4o-mini	Law Expert - History Expert	0.002 [-0.017, 0.020]	p = 0.850
GPT-4o-mini	Chemistry Expert - Physics Expert	0.002 [-0.016, 0.019]	p = 0.798
GPT-4o-mini	Chemistry Expert - History Expert	0.009 [-0.010, 0.027]	p = 0.373
GPT-4o-mini	Physics Expert - History Expert	0.006 [-0.013, 0.025]	p = 0.510
o3-mini	Baseline - Toddler	-0.050 [-0.070, -0.031]	p < 0.001
o3-mini	Baseline - Young Child	0.002 [-0.014, 0.018]	p = 0.838
o3-mini	Baseline - Layperson	-0.021 [-0.038, -0.003]	p = 0.022
o3-mini	Baseline - Engineering Expert	-0.004 [-0.019, 0.012]	p = 0.650
o3-mini	Baseline - Law Expert	0.010 [-0.007, 0.026]	p = 0.228
o3-mini	Baseline - Chemistry Expert	-0.006 [-0.023, 0.010]	p = 0.443
o3-mini	Baseline - Physics Expert	-0.004 [-0.020, 0.012]	p = 0.588

o3-mini	Baseline - History Expert	-0.003 [-0.020, 0.013]	p = 0.686
o3-mini	Toddler - Young Child	0.052 [0.034, 0.071]	p < 0.001
o3-mini	Toddler - Layperson	0.030 [0.010, 0.049]	p = 0.003
o3-mini	Toddler - Engineering Expert	0.047 [0.029, 0.065]	p < 0.001
o3-mini	Toddler - Law Expert	0.060 [0.040, 0.081]	p < 0.001
o3-mini	Toddler - Chemistry Expert	0.044 [0.025, 0.063]	p < 0.001
o3-mini	Toddler - Physics Expert	0.046 [0.028, 0.065]	p < 0.001
o3-mini	Toddler - History Expert	0.047 [0.029, 0.065]	p < 0.001
o3-mini	Young Child - Layperson	-0.023 [-0.039, -0.005]	p = 0.010
o3-mini	Young Child - Engineering Expert	-0.005 [-0.021, 0.011]	p = 0.525
o3-mini	Young Child - Law Expert	0.008 [-0.007, 0.024]	p = 0.298
o3-mini	Young Child - Chemistry Expert	-0.008 [-0.024, 0.008]	p = 0.339
o3-mini	Young Child - Physics Expert	-0.006 [-0.022, 0.010]	p = 0.483
o3-mini	Young Child - History Expert	-0.005 [-0.020, 0.011]	p = 0.530
o3-mini	Layperson - Engineering Expert	0.017 [-0.001, 0.035]	p = 0.059
o3-mini	Layperson - Law Expert	0.031 [0.013, 0.049]	p < 0.001
o3-mini	Layperson - Chemistry Expert	0.015 [-0.003, 0.032]	p = 0.108
o3-mini	Layperson - Physics Expert	0.017 [-0.000, 0.034]	p = 0.060
o3-mini	Layperson - History Expert	0.018 [-0.000, 0.035]	p = 0.054
o3-mini	Engineering Expert - Law Expert	0.014 [-0.003, 0.030]	p = 0.116
o3-mini	Engineering Expert - Chemistry Expert	-0.003 [-0.019, 0.014]	p = 0.745
o3-mini	Engineering Expert - Physics Expert	-0.001 [-0.017, 0.014]	p = 0.926
o3-mini	Engineering Expert - History Expert	0.000 [-0.016, 0.016]	p = 0.965
o3-mini	Law Expert - Chemistry Expert	-0.016 [-0.033, 0.001]	p = 0.059
o3-mini	Law Expert - Physics Expert	-0.014 [-0.032, 0.003]	p = 0.104
o3-mini	Law Expert - History Expert	-0.013 [-0.029, 0.002]	p = 0.096
o3-mini	Chemistry Expert - Physics Expert	0.002 [-0.014, 0.018]	p = 0.813
o3-mini	Chemistry Expert - History Expert	0.003 [-0.013, 0.019]	p = 0.713
o3-mini	Physics Expert - History Expert	0.001 [-0.016, 0.017]	p = 0.912
o4-mini	Baseline - Toddler	-0.047 [-0.069, -0.023]	p < 0.001
o4-mini	Baseline - Young Child	-0.004 [-0.023, 0.017]	p = 0.734
o4-mini	Baseline - Layperson	0.002 [-0.016, 0.020]	p = 0.865
o4-mini	Baseline - Engineering Expert	0.015 [-0.003, 0.033]	p = 0.093
o4-mini	Baseline - Law Expert	0.010 [-0.008, 0.028]	p = 0.284
o4-mini	Baseline - Chemistry Expert	0.010 [-0.008, 0.028]	p = 0.270

o4-mini	Baseline - Physics Expert	0.009 [-0.009, 0.027]	p = 0.308
o4-mini	Baseline - History Expert	0.002 [-0.016, 0.020]	p = 0.861
o4-mini	Toddler - Young Child	0.043 [0.023, 0.064]	p < 0.001
o4-mini	Toddler - Layperson	0.048 [0.026, 0.070]	p < 0.001
o4-mini	Toddler - Engineering Expert	0.062 [0.039, 0.083]	p < 0.001
o4-mini	Toddler - Law Expert	0.057 [0.034, 0.080]	p < 0.001
o4-mini	Toddler - Chemistry Expert	0.057 [0.035, 0.079]	p < 0.001
o4-mini	Toddler - Physics Expert	0.056 [0.034, 0.077]	p < 0.001
o4-mini	Toddler - History Expert	0.048 [0.026, 0.071]	p < 0.001
o4-mini	Young Child - Layperson	0.005 [-0.013, 0.024]	p = 0.566
o4-mini	Young Child - Engineering Expert	0.019 [0.000, 0.038]	p = 0.053
o4-mini	Young Child - Law Expert	0.013 [-0.006, 0.033]	p = 0.182
o4-mini	Young Child - Chemistry Expert	0.014 [-0.005, 0.033]	p = 0.157
o4-mini	Young Child - Physics Expert	0.013 [-0.006, 0.031]	p = 0.192
o4-mini	Young Child - History Expert	0.005 [-0.014, 0.024]	p = 0.603
o4-mini	Layperson - Engineering Expert	0.014 [-0.005, 0.031]	p = 0.147
o4-mini	Layperson - Law Expert	0.008 [-0.010, 0.026]	p = 0.384
o4-mini	Layperson - Chemistry Expert	0.009 [-0.009, 0.027]	p = 0.350
o4-mini	Layperson - Physics Expert	0.007 [-0.011, 0.026]	p = 0.427
o4-mini	Layperson - History Expert	-0.000 [-0.018, 0.018]	p = 0.990
o4-mini	Engineering Expert - Law Expert	-0.005 [-0.022, 0.012]	p = 0.527
o4-mini	Engineering Expert - Chemistry Expert	-0.005 [-0.022, 0.013]	p = 0.564
o4-mini	Engineering Expert - Physics Expert	-0.006 [-0.023, 0.010]	p = 0.461
o4-mini	Engineering Expert - History Expert	-0.014 [-0.031, 0.004]	p = 0.124
o4-mini	Law Expert - Chemistry Expert	0.000 [-0.016, 0.017]	p = 0.963
o4-mini	Law Expert - Physics Expert	-0.001 [-0.017, 0.016]	p = 0.923
o4-mini	Law Expert - History Expert	-0.008 [-0.025, 0.008]	p = 0.326
o4-mini	Chemistry Expert - Physics Expert	-0.001 [-0.018, 0.015]	p = 0.876
o4-mini	Chemistry Expert - History Expert	-0.009 [-0.026, 0.009]	p = 0.355
o4-mini	Physics Expert - History Expert	-0.007 [-0.025, 0.010]	p = 0.406
Gemini 2.0 Flash	Baseline - Toddler	-0.160 [-0.218, -0.099]	p < 0.001
Gemini 2.0 Flash	Baseline - Young Child	0.041 [-0.020, 0.102]	p = 0.193
Gemini 2.0 Flash	Baseline - Layperson	0.068 [0.010, 0.125]	p = 0.023
Gemini 2.0 Flash	Baseline - Engineering Expert	0.089 [0.033, 0.148]	p = 0.002
Gemini 2.0 Flash	Baseline - Law Expert	0.088 [0.031, 0.149]	p = 0.003

Gemini 2.0 Flash	Baseline - Chemistry Expert	0.068 [0.013, 0.125]	p = 0.017
Gemini 2.0 Flash	Baseline - Physics Expert	0.071 [0.015, 0.129]	p = 0.014
Gemini 2.0 Flash	Baseline - History Expert	0.077 [0.019, 0.136]	p = 0.009
Gemini 2.0 Flash	Toddler - Young Child	0.200 [0.141, 0.258]	p < 0.001
Gemini 2.0 Flash	Toddler - Layperson	0.228 [0.169, 0.289]	p < 0.001
Gemini 2.0 Flash	Toddler - Engineering Expert	0.235 [0.172, 0.298]	p < 0.001
Gemini 2.0 Flash	Toddler - Law Expert	0.242 [0.178, 0.305]	p < 0.001
Gemini 2.0 Flash	Toddler - Chemistry Expert	0.222 [0.158, 0.284]	p < 0.001
Gemini 2.0 Flash	Toddler - Physics Expert	0.225 [0.162, 0.288]	p < 0.001
Gemini 2.0 Flash	Toddler - History Expert	0.225 [0.162, 0.286]	p < 0.001
Gemini 2.0 Flash	Young Child - Layperson	0.027 [-0.029, 0.081]	p = 0.349
Gemini 2.0 Flash	Young Child - Engineering Expert	0.035 [-0.022, 0.095]	p = 0.245
Gemini 2.0 Flash	Young Child - Law Expert	0.042 [-0.017, 0.101]	p = 0.171
Gemini 2.0 Flash	Young Child - Chemistry Expert	0.022 [-0.039, 0.081]	p = 0.468
Gemini 2.0 Flash	Young Child - Physics Expert	0.025 [-0.034, 0.084]	p = 0.394
Gemini 2.0 Flash	Young Child - History Expert	0.030 [-0.030, 0.089]	p = 0.324
Gemini 2.0 Flash	Layperson - Engineering Expert	0.008 [-0.048, 0.065]	p = 0.785
Gemini 2.0 Flash	Layperson - Law Expert	0.013 [-0.044, 0.071]	p = 0.663
Gemini 2.0 Flash	Layperson - Chemistry Expert	-0.005 [-0.064, 0.054]	p = 0.860
Gemini 2.0 Flash	Layperson - Physics Expert	-0.002 [-0.061, 0.056]	p = 0.944
Gemini 2.0 Flash	Layperson - History Expert	-0.003 [-0.064, 0.056]	p = 0.915
Gemini 2.0 Flash	Engineering Expert - Law Expert	0.007 [-0.031, 0.047]	p = 0.720
Gemini 2.0 Flash	Engineering Expert - Chemistry Expert	-0.016 [-0.056, 0.022]	p = 0.417
Gemini 2.0 Flash	Engineering Expert - Physics Expert	-0.018 [-0.057, 0.022]	p = 0.366
Gemini 2.0 Flash	Engineering Expert - History Expert	-0.009 [-0.050, 0.033]	p = 0.692
Gemini 2.0 Flash	Law Expert - Chemistry Expert	-0.020 [-0.054, 0.015]	p = 0.265
Gemini 2.0 Flash	Law Expert - Physics Expert	-0.020 [-0.058, 0.018]	p = 0.290
Gemini 2.0 Flash	Law Expert - History Expert	-0.014 [-0.041, 0.015]	p = 0.338
Gemini 2.0 Flash	Chemistry Expert - Physics Expert	-0.000 [-0.038, 0.038]	p = 0.991
Gemini 2.0 Flash	Chemistry Expert - History Expert	0.006 [-0.032, 0.044]	p = 0.734
Gemini 2.0 Flash	Physics Expert - History Expert	0.007 [-0.030, 0.048]	p = 0.723
Gemini 2.5 Flash	Baseline - Toddler	-0.060 [-0.089, -0.031]	p < 0.001
Gemini 2.5 Flash	Baseline - Young Child	0.002 [-0.023, 0.028]	p = 0.867
Gemini 2.5 Flash	Baseline - Layperson	-0.025 [-0.048, -0.001]	p = 0.036
Gemini 2.5 Flash	Baseline - Engineering Expert	-0.054 [-0.074, -0.033]	p < 0.001

Gemini 2.5 Flash	Baseline - Law Expert	-0.225 [-0.265, -0.184]	p < 0.001
Gemini 2.5 Flash	Baseline - Chemistry Expert	-0.159 [-0.192, -0.125]	p < 0.001
Gemini 2.5 Flash	Baseline - Physics Expert	-0.101 [-0.128, -0.073]	p < 0.001
Gemini 2.5 Flash	Baseline - History Expert	-0.247 [-0.305, -0.188]	p < 0.001
Gemini 2.5 Flash	Toddler - Young Child	0.062 [0.039, 0.086]	p < 0.001
Gemini 2.5 Flash	Toddler - Layperson	0.035 [0.007, 0.062]	p = 0.011
Gemini 2.5 Flash	Toddler - Engineering Expert	0.006 [-0.023, 0.035]	p = 0.659
Gemini 2.5 Flash	Toddler - Law Expert	-0.165 [-0.205, -0.124]	p < 0.001
Gemini 2.5 Flash	Toddler - Chemistry Expert	-0.099 [-0.133, -0.066]	p < 0.001
Gemini 2.5 Flash	Toddler - Physics Expert	-0.041 [-0.072, -0.011]	p = 0.009
Gemini 2.5 Flash	Toddler - History Expert	-0.187 [-0.243, -0.132]	p < 0.001
Gemini 2.5 Flash	Young Child - Layperson	-0.027 [-0.052, -0.001]	p = 0.037
Gemini 2.5 Flash	Young Child - Engineering Expert	-0.056 [-0.084, -0.028]	p < 0.001
Gemini 2.5 Flash	Young Child - Law Expert	-0.228 [-0.271, -0.182]	p < 0.001
Gemini 2.5 Flash	Young Child - Chemistry Expert	-0.161 [-0.197, -0.128]	p < 0.001
Gemini 2.5 Flash	Young Child - Physics Expert	-0.103 [-0.134, -0.073]	p < 0.001
Gemini 2.5 Flash	Young Child - History Expert	-0.249 [-0.307, -0.192]	p < 0.001
Gemini 2.5 Flash	Layperson - Engineering Expert	-0.029 [-0.053, -0.005]	p = 0.015
Gemini 2.5 Flash	Layperson - Law Expert	-0.201 [-0.240, -0.162]	p < 0.001
Gemini 2.5 Flash	Layperson - Chemistry Expert	-0.134 [-0.169, -0.100]	p < 0.001
Gemini 2.5 Flash	Layperson - Physics Expert	-0.076 [-0.105, -0.047]	p < 0.001
Gemini 2.5 Flash	Layperson - History Expert	-0.222 [-0.281, -0.165]	p < 0.001
Gemini 2.5 Flash	Engineering Expert - Law Expert	-0.172 [-0.210, -0.133]	p < 0.001
Gemini 2.5 Flash	Engineering Expert - Chemistry Expert	-0.105 [-0.135, -0.075]	p < 0.001
Gemini 2.5 Flash	Engineering Expert - Physics Expert	-0.047 [-0.071, -0.022]	p < 0.001
Gemini 2.5 Flash	Engineering Expert - History Expert	-0.193 [-0.247, -0.137]	p < 0.001
Gemini 2.5 Flash	Law Expert - Chemistry Expert	0.066 [0.027, 0.107]	p = 0.001
Gemini 2.5 Flash	Law Expert - Physics Expert	0.124 [0.081, 0.167]	p < 0.001
Gemini 2.5 Flash	Law Expert - History Expert	-0.022 [-0.070, 0.026]	p = 0.358
Gemini 2.5 Flash	Chemistry Expert - Physics Expert	0.058 [0.031, 0.085]	p < 0.001
Gemini 2.5 Flash	Chemistry Expert - History Expert	-0.088 [-0.140, -0.036]	p = 0.001
Gemini 2.5 Flash	Physics Expert - History Expert	-0.146 [-0.201, -0.093]	p < 0.001

Note. N = 300 questions in each condition with 25 trials. Paired bootstrap-permutation tests (5,000 replicates). P-values represent the proportion of permuted differences with absolute

values exceeding observed differences under the null hypothesis. All p-values are uncorrected for multiple comparisons.

Table S4. Summary of adjacent and unrelated domain categorizations for domain-specific persona analysis.

Dataset-Domain Combination	Adjacent Expert	Unrelated Expert
GPQA Diamond - Physics	Math	Economics
GPQA Diamond - Chemistry	Physics	Economics
GPQA Diamond - Biology	Chemistry	Economics
MMLU-Pro - Engineering	Physics	Law
MMLU-Pro - Chemistry	Physics	Law
MMLU-Pro - Law	History	Engineering

Table S5. Pairwise mean comparison results between models and domain-related prompt variants on specific domains (GPQA Diamond).

Model (Domain)	Conditions	RD [95% CI]	Statistics
o4-mini (Physics)	Baseline - In-domain Expert	0.008 [-0.015, 0.031]	p = 0.531
o4-mini (Physics)	Baseline - Adjacent Expert	-0.003 [-0.027, 0.020]	p = 0.768
o4-mini (Physics)	Baseline - Unrelated Expert	-0.010 [-0.034, 0.013]	p = 0.408
o4-mini (Physics)	In-domain Expert - Adjacent Expert	-0.011 [-0.033, 0.011]	p = 0.338
o4-mini (Physics)	In-domain Expert - Unrelated Expert	-0.018 [-0.043, 0.007]	p = 0.174
o4-mini (Physics)	Adjacent Expert - Unrelated Expert	-0.007 [-0.031, 0.018]	p = 0.595
o3-mini (Physics)	Baseline - In-domain Expert	-0.007 [-0.031, 0.017]	p = 0.571
o3-mini (Physics)	Baseline - Adjacent Expert	-0.007 [-0.030, 0.016]	p = 0.565
o3-mini (Physics)	Baseline - Unrelated Expert	-0.015 [-0.040, 0.009]	p = 0.219
o3-mini (Physics)	In-domain Expert - Adjacent Expert	0.000 [-0.024, 0.024]	p = 0.995
o3-mini (Physics)	In-domain Expert - Unrelated Expert	-0.008 [-0.034, 0.017]	p = 0.522
o3-mini (Physics)	Adjacent Expert - Unrelated Expert	-0.008 [-0.033, 0.015]	p = 0.513
Gemini 2.5 Flash (Physics)	Baseline - In-domain Expert	-0.039 [-0.083, 0.004]	p = 0.077
Gemini 2.5 Flash (Physics)	Baseline - Adjacent Expert	0.018 [-0.025, 0.062]	p = 0.415
Gemini 2.5 Flash (Physics)	Baseline - Unrelated Expert	-0.247 [-0.324, -0.173]	p < 0.001
Gemini 2.5 Flash (Physics)	In-domain Expert - Adjacent Expert	0.057 [0.010, 0.106]	p = 0.024
Gemini 2.5 Flash (Physics)	In-domain Expert - Unrelated Expert	-0.208 [-0.278, -0.139]	p < 0.001

Gemini 2.5 Flash (Physics)	Adjacent Expert - Unrelated Expert	-0.265 [-0.346, -0.184]	p < 0.001
Gemini 2.0 Flash (Physics)	Baseline - In-domain Expert	0.018 [-0.094, 0.130]	p = 0.753
Gemini 2.0 Flash (Physics)	Baseline - Adjacent Expert	0.049 [-0.065, 0.161]	p = 0.402
Gemini 2.0 Flash (Physics)	Baseline - Unrelated Expert	-0.037 [-0.148, 0.077]	p = 0.516
Gemini 2.0 Flash (Physics)	In-domain Expert - Adjacent Expert	0.032 [-0.009, 0.073]	p = 0.134
Gemini 2.0 Flash (Physics)	In-domain Expert - Unrelated Expert	-0.063 [-0.115, -0.011]	p = 0.018
Gemini 2.0 Flash (Physics)	Adjacent Expert - Unrelated Expert	-0.095 [-0.151, -0.038]	p < 0.001
GPT-4o (Physics)	Baseline - In-domain Expert	-0.020 [-0.149, 0.116]	p = 0.776
GPT-4o (Physics)	Baseline - Adjacent Expert	-0.075 [-0.214, 0.060]	p = 0.275
GPT-4o (Physics)	Baseline - Unrelated Expert	-0.039 [-0.089, 0.011]	p = 0.122
GPT-4o (Physics)	In-domain Expert - Adjacent Expert	0.091 [-0.058, 0.239]	p = 0.236
GPT-4o (Physics)	In-domain Expert - Unrelated Expert	-0.001 [-0.140, 0.139]	p = 0.993
GPT-4o (Physics)	Adjacent Expert - Unrelated Expert	0.052 [-0.089, 0.198]	p = 0.490
GPT-4o-mini (Physics)	Baseline - In-domain Expert	0.013 [-0.026, 0.053]	p = 0.537
GPT-4o-mini (Physics)	Baseline - Adjacent Expert	0.005 [-0.041, 0.050]	p = 0.845
GPT-4o-mini (Physics)	Baseline - Unrelated Expert	-0.018 [-0.057, 0.022]	p = 0.374
GPT-4o-mini (Physics)	In-domain Expert - Adjacent Expert	-0.008 [-0.047, 0.031]	p = 0.679
GPT-4o-mini (Physics)	In-domain Expert - Unrelated Expert	-0.030 [-0.074, 0.014]	p = 0.174
GPT-4o-mini (Physics)	Adjacent Expert - Unrelated Expert	-0.022 [-0.070, 0.025]	p = 0.375
o4-mini (Chemistry)	Baseline - In-domain Expert	-0.016 [-0.049, 0.017]	p = 0.355
o4-mini (Chemistry)	Baseline - Adjacent Expert	-0.020 [-0.054, 0.013]	p = 0.244
o4-mini (Chemistry)	Baseline - Unrelated Expert	-0.030 [-0.062, 0.002]	p = 0.067
o4-mini (Chemistry)	In-domain Expert - Adjacent Expert	-0.004 [-0.037, 0.028]	p = 0.815
o4-mini (Chemistry)	In-domain Expert - Unrelated Expert	-0.014 [-0.047, 0.020]	p = 0.413
o4-mini (Chemistry)	Adjacent Expert - Unrelated Expert	-0.010 [-0.043, 0.022]	p = 0.535
o3-mini (Chemistry)	Baseline - In-domain Expert	-0.003 [-0.035, 0.030]	p = 0.883
o3-mini (Chemistry)	Baseline - Adjacent Expert	-0.007 [-0.039, 0.025]	p = 0.639
o3-mini (Chemistry)	Baseline - Unrelated Expert	-0.028 [-0.062, 0.006]	p = 0.110
o3-mini (Chemistry)	In-domain Expert - Adjacent Expert	-0.005 [-0.035, 0.026]	p = 0.747
o3-mini (Chemistry)	In-domain Expert - Unrelated Expert	-0.025 [-0.060, 0.010]	p = 0.168
o3-mini (Chemistry)	Adjacent Expert - Unrelated Expert	-0.020 [-0.052, 0.011]	p = 0.200
Gemini 2.5 Flash (Chemistry)	Baseline - In-domain Expert	0.005 [-0.028, 0.040]	p = 0.787
Gemini 2.5 Flash (Chemistry)	Baseline - Adjacent Expert	0.018 [-0.019, 0.055]	p = 0.334
Gemini 2.5 Flash	Baseline - Unrelated Expert	-0.084 [-0.133, -0.037]	p < 0.001

(Chemistry)			
Gemini 2.5 Flash (Chemistry)	In-domain Expert - Adjacent Expert	0.013 [-0.020, 0.047]	p = 0.445
Gemini 2.5 Flash (Chemistry)	In-domain Expert - Unrelated Expert	-0.089 [-0.139, -0.041]	p < 0.001
Gemini 2.5 Flash (Chemistry)	Adjacent Expert - Unrelated Expert	-0.102 [-0.154, -0.053]	p < 0.001
Gemini 2.0 Flash (Chemistry)	Baseline - In-domain Expert	-0.003 [-0.096, 0.088]	p = 0.941
Gemini 2.0 Flash (Chemistry)	Baseline - Adjacent Expert	0.007 [-0.084, 0.100]	p = 0.875
Gemini 2.0 Flash (Chemistry)	Baseline - Unrelated Expert	-0.020 [-0.111, 0.073]	p = 0.670
Gemini 2.0 Flash (Chemistry)	In-domain Expert - Adjacent Expert	0.002 [-0.030, 0.033]	p = 0.904
Gemini 2.0 Flash (Chemistry)	In-domain Expert - Unrelated Expert	-0.045 [-0.090, -0.000]	p = 0.049
Gemini 2.0 Flash (Chemistry)	Adjacent Expert - Unrelated Expert	-0.047 [-0.091, -0.002]	p = 0.039
GPT-4o (Chemistry)	Baseline - In-domain Expert	0.003 [-0.034, 0.042]	p = 0.866
GPT-4o (Chemistry)	Baseline - Adjacent Expert	-0.046 [-0.167, 0.070]	p = 0.442
GPT-4o (Chemistry)	Baseline - Unrelated Expert	-0.011 [-0.048, 0.027]	p = 0.569
GPT-4o (Chemistry)	In-domain Expert - Adjacent Expert	-0.047 [-0.170, 0.073]	p = 0.444
GPT-4o (Chemistry)	In-domain Expert - Unrelated Expert	-0.014 [-0.049, 0.022]	p = 0.480
GPT-4o (Chemistry)	Adjacent Expert - Unrelated Expert	0.039 [-0.084, 0.161]	p = 0.534
GPT-4o-mini (Chemistry)	Baseline - In-domain Expert	0.005 [-0.036, 0.046]	p = 0.813
GPT-4o-mini (Chemistry)	Baseline - Adjacent Expert	-0.021 [-0.056, 0.017]	p = 0.268
GPT-4o-mini (Chemistry)	Baseline - Unrelated Expert	-0.007 [-0.040, 0.026]	p = 0.674
GPT-4o-mini (Chemistry)	In-domain Expert - Adjacent Expert	-0.026 [-0.067, 0.016]	p = 0.228
GPT-4o-mini (Chemistry)	In-domain Expert - Unrelated Expert	-0.012 [-0.055, 0.031]	p = 0.574
GPT-4o-mini (Chemistry)	Adjacent Expert - Unrelated Expert	0.013 [-0.021, 0.047]	p = 0.427
o4-mini (Biology)	Baseline - In-domain Expert	0.011 [-0.042, 0.065]	p = 0.737
o4-mini (Biology)	Baseline - Adjacent Expert	0.013 [-0.040, 0.067]	p = 0.631
o4-mini (Biology)	Baseline - Unrelated Expert	0.011 [-0.040, 0.063]	p = 0.718
o4-mini (Biology)	In-domain Expert - Adjacent Expert	0.002 [-0.048, 0.053]	p = 0.900
o4-mini (Biology)	In-domain Expert - Unrelated Expert	0.000 [-0.055, 0.055]	p = 1.000
o4-mini (Biology)	Adjacent Expert - Unrelated Expert	-0.002 [-0.053, 0.051]	p = 0.906
o3-mini (Biology)	Baseline - In-domain Expert	-0.004 [-0.061, 0.055]	p = 0.917
o3-mini (Biology)	Baseline - Adjacent Expert	0.000 [-0.065, 0.067]	p = 1.000

o3-mini (Biology)	Baseline - Unrelated Expert	0.015 [-0.055, 0.084]	p = 0.653
o3-mini (Biology)	In-domain Expert - Adjacent Expert	0.004 [-0.055, 0.063]	p = 0.924
o3-mini (Biology)	In-domain Expert - Unrelated Expert	0.019 [-0.042, 0.080]	p = 0.582
o3-mini (Biology)	Adjacent Expert - Unrelated Expert	0.015 [-0.042, 0.072]	p = 0.580
Gemini 2.5 Flash (Biology)	Baseline - In-domain Expert	0.004 [-0.073, 0.082]	p = 0.925
Gemini 2.5 Flash (Biology)	Baseline - Adjacent Expert	0.000 [-0.079, 0.082]	p = 0.993
Gemini 2.5 Flash (Biology)	Baseline - Unrelated Expert	-0.162 [-0.280, -0.044]	p = 0.007
Gemini 2.5 Flash (Biology)	In-domain Expert - Adjacent Expert	-0.004 [-0.084, 0.076]	p = 0.936
Gemini 2.5 Flash (Biology)	In-domain Expert - Unrelated Expert	-0.166 [-0.284, -0.051]	p = 0.003
Gemini 2.5 Flash (Biology)	Adjacent Expert - Unrelated Expert	-0.162 [-0.290, -0.034]	p = 0.009
Gemini 2.0 Flash (Biology)	Baseline - In-domain Expert	0.086 [-0.105, 0.274]	p = 0.413
Gemini 2.0 Flash (Biology)	Baseline - Adjacent Expert	0.112 [-0.083, 0.317]	p = 0.295
Gemini 2.0 Flash (Biology)	Baseline - Unrelated Expert	0.126 [-0.055, 0.311]	p = 0.207
Gemini 2.0 Flash (Biology)	In-domain Expert - Adjacent Expert	0.029 [-0.038, 0.094]	p = 0.404
Gemini 2.0 Flash (Biology)	In-domain Expert - Unrelated Expert	0.040 [-0.040, 0.120]	p = 0.320
Gemini 2.0 Flash (Biology)	Adjacent Expert - Unrelated Expert	0.011 [-0.061, 0.084]	p = 0.776
GPT-4o (Biology)	Baseline - In-domain Expert	-0.008 [-0.076, 0.059]	p = 0.829
GPT-4o (Biology)	Baseline - Adjacent Expert	0.029 [-0.036, 0.097]	p = 0.400
GPT-4o (Biology)	Baseline - Unrelated Expert	-0.017 [-0.080, 0.044]	p = 0.637
GPT-4o (Biology)	In-domain Expert - Adjacent Expert	0.038 [-0.019, 0.097]	p = 0.243
GPT-4o (Biology)	In-domain Expert - Unrelated Expert	-0.008 [-0.072, 0.051]	p = 0.759
GPT-4o (Biology)	Adjacent Expert - Unrelated Expert	-0.046 [-0.109, 0.017]	p = 0.159
GPT-4o-mini (Biology)	Baseline - In-domain Expert	-0.004 [-0.063, 0.053]	p = 0.921
GPT-4o-mini (Biology)	Baseline - Adjacent Expert	0.021 [-0.048, 0.088]	p = 0.547
GPT-4o-mini (Biology)	Baseline - Unrelated Expert	0.011 [-0.046, 0.067]	p = 0.706
GPT-4o-mini (Biology)	In-domain Expert - Adjacent Expert	0.025 [-0.027, 0.078]	p = 0.342
GPT-4o-mini (Biology)	In-domain Expert - Unrelated Expert	0.015 [-0.036, 0.063]	p = 0.564
GPT-4o-mini (Biology)	Adjacent Expert - Unrelated Expert	-0.011 [-0.072, 0.051]	p = 0.730

Table S6. Pairwise mean comparison results between models and domain-related prompt variants on specific domains (MMLU-Pro).

Model (Domain)	Conditions	RD [95% CI]	Statistics
o4-mini (Engineering)	Baseline - In-domain Expert	0.002 [-0.027, 0.032]	p = 0.897
o4-mini (Engineering)	Baseline - Adjacent Expert	-0.013 [-0.042, 0.015]	p = 0.361

o4-mini (Engineering)	Baseline - Unrelated Expert	-0.016 [-0.045, 0.013]	p = 0.278
o4-mini (Engineering)	In-domain Expert - Adjacent Expert	-0.015 [-0.044, 0.013]	p = 0.302
o4-mini (Engineering)	In-domain Expert - Unrelated Expert	-0.018 [-0.046, 0.010]	p = 0.221
o4-mini (Engineering)	Adjacent Expert - Unrelated Expert	-0.003 [-0.033, 0.026]	p = 0.843
o3-mini (Engineering)	Baseline - In-domain Expert	0.007 [-0.021, 0.036]	p = 0.628
o3-mini (Engineering)	Baseline - Adjacent Expert	0.024 [-0.005, 0.052]	p = 0.107
o3-mini (Engineering)	Baseline - Unrelated Expert	0.018 [-0.012, 0.049]	p = 0.250
o3-mini (Engineering)	In-domain Expert - Adjacent Expert	0.016 [-0.013, 0.045]	p = 0.270
o3-mini (Engineering)	In-domain Expert - Unrelated Expert	0.011 [-0.021, 0.042]	p = 0.488
o3-mini (Engineering)	Adjacent Expert - Unrelated Expert	-0.005 [-0.036, 0.026]	p = 0.753
Gemini 2.5 Flash (Engineering)	Baseline - In-domain Expert	-0.033 [-0.064, -0.003]	p = 0.034
Gemini 2.5 Flash (Engineering)	Baseline - Adjacent Expert	-0.013 [-0.044, 0.017]	p = 0.398
Gemini 2.5 Flash (Engineering)	Baseline - Unrelated Expert	-0.305 [-0.387, -0.222]	p < 0.001
Gemini 2.5 Flash (Engineering)	In-domain Expert - Adjacent Expert	0.020 [-0.010, 0.051]	p = 0.194
Gemini 2.5 Flash (Engineering)	In-domain Expert - Unrelated Expert	-0.271 [-0.345, -0.197]	p < 0.001
Gemini 2.5 Flash (Engineering)	Adjacent Expert - Unrelated Expert	-0.291 [-0.372, -0.209]	p < 0.001
Gemini 2.0 Flash (Engineering)	Baseline - In-domain Expert	0.090 [0.029, 0.152]	p = 0.003
Gemini 2.0 Flash (Engineering)	Baseline - Adjacent Expert	0.071 [0.010, 0.134]	p = 0.024
Gemini 2.0 Flash (Engineering)	Baseline - Unrelated Expert	0.082 [0.019, 0.149]	p = 0.009
Gemini 2.0 Flash (Engineering)	In-domain Expert - Adjacent Expert	-0.017 [-0.048, 0.016]	p = 0.328
Gemini 2.0 Flash (Engineering)	In-domain Expert - Unrelated Expert	-0.008 [-0.041, 0.024]	p = 0.623
Gemini 2.0 Flash (Engineering)	Adjacent Expert - Unrelated Expert	0.008 [-0.026, 0.042]	p = 0.636
GPT-4o (Engineering)	Baseline - In-domain Expert	-0.009 [-0.054, 0.035]	p = 0.714
GPT-4o (Engineering)	Baseline - Adjacent Expert	-0.008 [-0.102, 0.083]	p = 0.873
GPT-4o (Engineering)	Baseline - Unrelated Expert	-0.060 [-0.095, -0.024]	p = 0.001
GPT-4o (Engineering)	In-domain Expert - Adjacent Expert	-0.003 [-0.094, 0.089]	p = 0.955
GPT-4o (Engineering)	In-domain Expert - Unrelated Expert	-0.051 [-0.097, -0.007]	p = 0.026
GPT-4o (Engineering)	Adjacent Expert - Unrelated Expert	-0.053 [-0.153, 0.042]	p = 0.278

GPT-4o-mini (Engineering)	Baseline - In-domain Expert	0.013 [-0.055, 0.081]	p = 0.702
GPT-4o-mini (Engineering)	Baseline - Adjacent Expert	-0.004 [-0.037, 0.029]	p = 0.798
GPT-4o-mini (Engineering)	Baseline - Unrelated Expert	-0.031 [-0.067, 0.003]	p = 0.079
GPT-4o-mini (Engineering)	In-domain Expert - Adjacent Expert	-0.019 [-0.088, 0.049]	p = 0.590
GPT-4o-mini (Engineering)	In-domain Expert - Unrelated Expert	-0.046 [-0.113, 0.021]	p = 0.184
GPT-4o-mini (Engineering)	Adjacent Expert - Unrelated Expert	-0.027 [-0.061, 0.007]	p = 0.124
o4-mini (Chemistry)	Baseline - In-domain Expert	0.010 [-0.015, 0.036]	p = 0.449
o4-mini (Chemistry)	Baseline - Adjacent Expert	0.006 [-0.019, 0.030]	p = 0.656
o4-mini (Chemistry)	Baseline - Unrelated Expert	0.009 [-0.019, 0.037]	p = 0.554
o4-mini (Chemistry)	In-domain Expert - Adjacent Expert	-0.005 [-0.029, 0.021]	p = 0.727
o4-mini (Chemistry)	In-domain Expert - Unrelated Expert	-0.001 [-0.027, 0.024]	p = 0.914
o4-mini (Chemistry)	Adjacent Expert - Unrelated Expert	0.003 [-0.022, 0.028]	p = 0.806
o3-mini (Chemistry)	Baseline - In-domain Expert	-0.010 [-0.036, 0.015]	p = 0.446
o3-mini (Chemistry)	Baseline - Adjacent Expert	-0.015 [-0.039, 0.009]	p = 0.228
o3-mini (Chemistry)	Baseline - Unrelated Expert	0.008 [-0.017, 0.033]	p = 0.519
o3-mini (Chemistry)	In-domain Expert - Adjacent Expert	-0.005 [-0.030, 0.020]	p = 0.731
o3-mini (Chemistry)	In-domain Expert - Unrelated Expert	0.018 [-0.009, 0.046]	p = 0.199
o3-mini (Chemistry)	Adjacent Expert - Unrelated Expert	0.023 [-0.002, 0.049]	p = 0.073
Gemini 2.5 Flash (Chemistry)	Baseline - In-domain Expert	-0.052 [-0.092, -0.014]	p = 0.008
Gemini 2.5 Flash (Chemistry)	Baseline - Adjacent Expert	-0.051 [-0.090, -0.013]	p = 0.009
Gemini 2.5 Flash (Chemistry)	Baseline - Unrelated Expert	-0.359 [-0.446, -0.272]	p < 0.001
Gemini 2.5 Flash (Chemistry)	In-domain Expert - Adjacent Expert	0.001 [-0.034, 0.036]	p = 0.950
Gemini 2.5 Flash (Chemistry)	In-domain Expert - Unrelated Expert	-0.307 [-0.386, -0.227]	p < 0.001
Gemini 2.5 Flash (Chemistry)	Adjacent Expert - Unrelated Expert	-0.308 [-0.387, -0.229]	p < 0.001
Gemini 2.0 Flash (Chemistry)	Baseline - In-domain Expert	0.149 [0.051, 0.249]	p = 0.002
Gemini 2.0 Flash (Chemistry)	Baseline - Adjacent Expert	0.131 [0.029, 0.233]	p = 0.010
Gemini 2.0 Flash (Chemistry)	Baseline - Unrelated Expert	0.236 [0.119, 0.353]	p < 0.001
Gemini 2.0 Flash (Chemistry)	In-domain Expert - Adjacent Expert	-0.020 [-0.130, 0.088]	p = 0.703
Gemini 2.0 Flash (Chemistry)	In-domain Expert - Unrelated Expert	0.077 [-0.022, 0.177]	p = 0.138

Gemini 2.0 Flash (Chemistry)	Adjacent Expert - Unrelated Expert	0.107 [-0.006, 0.217]	p = 0.064
GPT-4o (Chemistry)	Baseline - In-domain Expert	-0.003 [-0.106, 0.096]	p = 0.949
GPT-4o (Chemistry)	Baseline - Adjacent Expert	-0.004 [-0.102, 0.094]	p = 0.930
GPT-4o (Chemistry)	Baseline - Unrelated Expert	-0.022 [-0.050, 0.005]	p = 0.124
GPT-4o (Chemistry)	In-domain Expert - Adjacent Expert	0.002 [-0.090, 0.103]	p = 0.968
GPT-4o (Chemistry)	In-domain Expert - Unrelated Expert	-0.018 [-0.117, 0.083]	p = 0.728
GPT-4o (Chemistry)	Adjacent Expert - Unrelated Expert	-0.018 [-0.122, 0.084]	p = 0.746
GPT-4o-mini (Chemistry)	Baseline - In-domain Expert	-0.003 [-0.031, 0.025]	p = 0.845
GPT-4o-mini (Chemistry)	Baseline - Adjacent Expert	0.007 [-0.022, 0.036]	p = 0.627
GPT-4o-mini (Chemistry)	Baseline - Unrelated Expert	0.017 [-0.013, 0.047]	p = 0.259
GPT-4o-mini (Chemistry)	In-domain Expert - Adjacent Expert	0.010 [-0.018, 0.039]	p = 0.485
GPT-4o-mini (Chemistry)	In-domain Expert - Unrelated Expert	0.020 [-0.010, 0.050]	p = 0.189
GPT-4o-mini (Chemistry)	Adjacent Expert - Unrelated Expert	0.010 [-0.020, 0.040]	p = 0.521
o4-mini (Law)	Baseline - In-domain Expert	0.037 [0.000, 0.074]	p = 0.049
o4-mini (Law)	Baseline - Adjacent Expert	0.021 [-0.014, 0.056]	p = 0.243
o4-mini (Law)	Baseline - Unrelated Expert	0.032 [-0.004, 0.070]	p = 0.090
o4-mini (Law)	In-domain Expert - Adjacent Expert	-0.016 [-0.050, 0.018]	p = 0.361
o4-mini (Law)	In-domain Expert - Unrelated Expert	-0.005 [-0.042, 0.031]	p = 0.798
o4-mini (Law)	Adjacent Expert - Unrelated Expert	0.011 [-0.023, 0.046]	p = 0.519
o3-mini (Law)	Baseline - In-domain Expert	0.003 [-0.026, 0.033]	p = 0.839
o3-mini (Law)	Baseline - Adjacent Expert	-0.019 [-0.048, 0.010]	p = 0.203
o3-mini (Law)	Baseline - Unrelated Expert	-0.005 [-0.033, 0.022]	p = 0.699
o3-mini (Law)	In-domain Expert - Adjacent Expert	-0.022 [-0.051, 0.007]	p = 0.136
o3-mini (Law)	In-domain Expert - Unrelated Expert	-0.008 [-0.038, 0.021]	p = 0.566
o3-mini (Law)	Adjacent Expert - Unrelated Expert	0.013 [-0.012, 0.039]	p = 0.316
Gemini 2.5 Flash (Law)	Baseline - In-domain Expert	-0.013 [-0.047, 0.021]	p = 0.464
Gemini 2.5 Flash (Law)	Baseline - Adjacent Expert	-0.066 [-0.175, 0.040]	p = 0.238
Gemini 2.5 Flash (Law)	Baseline - Unrelated Expert	-0.063 [-0.101, -0.026]	p = 0.001
Gemini 2.5 Flash (Law)	In-domain Expert - Adjacent Expert	-0.055 [-0.161, 0.050]	p = 0.308
Gemini 2.5 Flash (Law)	In-domain Expert - Unrelated Expert	-0.051 [-0.088, -0.013]	p = 0.008
Gemini 2.5 Flash (Law)	Adjacent Expert - Unrelated Expert	0.004 [-0.103, 0.112]	p = 0.945
Gemini 2.0 Flash (Law)	Baseline - In-domain Expert	-0.022 [-0.148, 0.097]	p = 0.717
Gemini 2.0 Flash (Law)	Baseline - Adjacent Expert	-0.029 [-0.156, 0.094]	p = 0.648
Gemini 2.0 Flash (Law)	Baseline - Unrelated Expert	0.009 [-0.109, 0.132]	p = 0.883

Gemini 2.0 Flash (Law)	In-domain Expert - Adjacent Expert	-0.007 [-0.036, 0.022]	p = 0.635
Gemini 2.0 Flash (Law)	In-domain Expert - Unrelated Expert	0.008 [-0.064, 0.077]	p = 0.835
Gemini 2.0 Flash (Law)	Adjacent Expert - Unrelated Expert	0.013 [-0.055, 0.081]	p = 0.720
GPT-4o (Law)	Baseline - In-domain Expert	0.042 [0.008, 0.076]	p = 0.014
GPT-4o (Law)	Baseline - Adjacent Expert	0.007 [-0.107, 0.117]	p = 0.912
GPT-4o (Law)	Baseline - Unrelated Expert	0.006 [-0.107, 0.118]	p = 0.911
GPT-4o (Law)	In-domain Expert - Adjacent Expert	-0.038 [-0.147, 0.073]	p = 0.506
GPT-4o (Law)	In-domain Expert - Unrelated Expert	-0.038 [-0.153, 0.074]	p = 0.498
GPT-4o (Law)	Adjacent Expert - Unrelated Expert	-0.003 [-0.032, 0.026]	p = 0.846
GPT-4o-mini (Law)	Baseline - In-domain Expert	0.014 [-0.020, 0.047]	p = 0.424
GPT-4o-mini (Law)	Baseline - Adjacent Expert	-0.017 [-0.050, 0.015]	p = 0.305
GPT-4o-mini (Law)	Baseline - Unrelated Expert	-0.014 [-0.121, 0.092]	p = 0.818
GPT-4o-mini (Law)	In-domain Expert - Adjacent Expert	-0.032 [-0.063, 0.000]	p = 0.053
GPT-4o-mini (Law)	In-domain Expert - Unrelated Expert	-0.027 [-0.136, 0.078]	p = 0.613
GPT-4o-mini (Law)	Adjacent Expert - Unrelated Expert	0.004 [-0.103, 0.106]	p = 0.937
